F-44 FALCON

PDR Report

AE460A 2025 AIAA Stike Fighter Aircraft

Preliminary Design review

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ALEBORNE - F-44 Falcon

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1 Executive Summary

1.1 Description

Modern naval aviation faces a rapidly evolving threat landscape defined by long-range anti-ship missiles, advanced integrated air defense systems, and highly maneuverable fifth-generation adversary aircraft. To preserve carrier air wing dominance through 2035 and beyond, the next generation of carrier-based strike fighters must deliver greater reach, survivability, and flexibility. It does this without driving unit acquisition cost beyond sustainable limits. The 2025–2026 AIAA Navy Request for Proposal challenges designers to break this cycle of escalating cost while meaningfully improving capability over the F/A-18E/F Super Hornet. Aleborne is our response to that challenge: a highly capable, multi-mission strike fighter engineered from the outset to excel in both air-to-air combat and long-range precision strike, while remaining fully compatible with CVN-68 and CVN-78 class aircraft carriers.

Aleborne's design integrates three core principles: range, survivability, and mission flexibility. A carefully optimized aerodynamic configuration combined with fuel-efficient engines allows the aircraft to meet the required 700 nmi combat radius, enabling it to operate well beyond the reach of current carrier-based assets. Internal carriage of primary weapons improves survivability through reduced radar cross-section, while also meeting the requirement for carrying all ordnance throughout the mission without jettison. The design incorporates materials suited for a 25-year maritime service life and all subsystems meet or exceed Technology Readiness Level 6, ensuring both durability and practical manufacturability.

In its air-to-air mission, Aleborne carries six AIM-120C missiles and two AIM-9X missiles, giving it the firepower and agility required to defend the carrier strike group against modern enemy fighters. With a maximum dash speed of Mach 1.49 at 30,000 ft, Aleborne ensures the carrier air wing retains the ability to rapidly intercept high-speed threats. We are still a little shy of the required speed, but work is underway to improve than and to further improve instantaneous turn performance and high-Mach energy retention as the aerodynamic shaping matures.

For the strike mission, Aleborne delivers four Mk-83 JDAMs and two AIM-9X missiles with precision and reliability. Its sea-level dash requirement of Mach 0.85 is currently in progress, with aerodynamic refinements focused on reducing transonic drag and improving low-altitude efficiency. The design comfortably meets carrier suitability requirements, including launch from the C-13-2 catapult and recovery with the required

arresting gear margins, while staying within the dimensional limits for flight deck operations and hangar stowage.

Aleborne achieves full compliance with the majority of system-level requirements at this stage of development. General design, avionics and payload capacity, carrier suitability, and recovery performance are all satisfied. Mission-level parameters including combat radius and weapons carriage have been validated. Remaining performance elements, in particular sustained turn rate, combat maneuvering time, low-altitude dash speed, and high-altitude dash capability, are well within reach and are the focus of ongoing aerodynamic and propulsion optimization. The current design trajectory indicates a clear path to meeting all required thresholds.

Aleborne represents a balanced, forward-leaning solution to the Navy's next-generation strike fighter needs. It combines proven systems, carrier readiness, and advanced mission capability in a design optimized for both performance and cost sustainability. As the design matures, Aleborne is positioned to deliver a leap in capability while maintaining the operational practicality and versatility that have long defined the Navy's most successful carrier-based aircraft.

1.1.2 Operational View (OV-1)

The requirements for the Next Generation Strike Fighter Aircraft, as defined by the AIAA Request for Proposal, establish the foundational capabilities the aircraft must achieve. These requirements are organized into three categories: general system-level requirements, air-to-air combat mission requirements, and strike mission requirements. The general requirements are summarized in Table 1, the air-to-air mission requirements in Table 2, the strike mission requirements in Table 3, and design features in Table 4. The aircraft is required to successfully execute both missions while carrying the full specified payload.

Table 1-1. Key Requirements

Key Requirements	Requirement Type
Cost:	
Materials: Materials shall be suitable for a 25-year service life in a maritime environment and must be currently available	Required
Subsystems: All subsystems technologies shall currently be at technology	Required

Key Requirements	Requirement
readiness level 6 or higher.	
Ordnance: External Carriage of Ordnance	Required
Ordnance: Internal carriage of weapons for reduced radar cross section	Desired
Carrier Suitability: The aircraft shall be operable from CVN-68 and CVN-78 class aircraft carriers with minimal impact on operations for the carrier air wing.	Required
Launch Performance: The aircraft shall be capable of launch on a tropical day with zero wind over the deck	Required
Recovery Performance: Arrestment engaging speed shall be assumed to be 5% greater than approach speed. Approach airspeed shall be greater than 10% above stall speed, but less than 145 knots. Arrestment landing weight shall include sufficient fuel for 20 minutes loiter at 10,000 ft and two landing attempts, 25% maximum fuel weight, and 50% store weight.	Required
Payload: Avionics/sensor weight shall be 2500 lb	Required
Point Performance: The design vertical load factor (Nz) shall be greater than 7g at mid mission weight.	Required
Point Performance: The design vertical load factor (Nz) shall be greater than 8.0g at mid mission weight	Desired
Point Performance: Aircraft external store carriage capability shall not be less than 10,000 lb in total or less than 3,000 lb for any single store.	Required
Point Performance: Launch requirements must be achievable with a Wind Over Deck (WOD) no greater than zero and using the C-13-2 catapult	Required
Point Performance: Recovery requirements must be achievable with a WOD not greater than 15 knots and using the Mark 7 Mod 3 arresting gear	Required
Point Performance: If multi-engine, launch single-engine rate of climb (SEROC) not less than 200 ft/min	Required
Point Performance: If multi-engine, approach SEROC not less than 500 ft/min	Required
Point Performance: Unfolded wingspan shall not exceed 60 feet.	Required
Point Performance: Folded wingspan shall not exceed 35 feet	Required
Point Performance: Overall aircraft length shall not exceed 50 feet	Required
Point Performance: Overall aircraft height shall not exceed 18.5 feet	Required
Point Performance: Maximum take-off gross weight shall not be greater than 90,000 lb	Required

Table 1-2. A2A Combat Requirements

Air to Air Combat Requirements	Requirement Type
A 700 nm. minimum combat radius	Required
A 1,000 nm. combat radius	Desired
Combat shall consist of a minimum of 2 minutes at maximum thrust at best turning speed at 10,000 ft.	Required
A combat time of 5 minutes at maximum thrust at best turning speed at 10,000 ft	Desired
Ordinance shall be carried for the entire mission	Required
The aircraft shall have an 8.0 deg/sec minimum sustained turn rate at 20,000 ft at the mid mission fuel weight	Required
The aircraft shall have an 10.0 deg/sec minimum sustained turn rate at 20,000 ft at the mid mission fuel weight	Desired
A Mach 1.6 dash speed at 30,000 ft	Required
Mach 2.0 dash speed at 30,000 ft	Desired
Ordnance load shall be 6 “AIM-120C” & 2 “AIM-9X”	Required

Table 1-3. Strike Mission Requirements

Strike Mission Requirements	Requirement type
A 700 nm. minimum combat radius	Required
A 1,000 nm. combat radius	Desired
Mid mission combat shall consist of 50 nm sea level dashes at intermediate thrust	Required
Sea level dash speed shall be Mach .85 or greater (intermediate thrust)	Required
Sea level dash speed shall be Mach .90 or greater (intermediate thrust)	Desired
Ordinance shall be carried for the entire mission (including arrestment).	Required
Ordnance load shall be 4 “MK-83 JDAM” & 2 “AIM-9X”	Required

Table 1-4. Design Features

Category	Design Feature	Reason
Airframe Layout	Large wing area	Internal fuel capacity and range
Airframe Layout	Low Taper Ratio	Reduces wing bending moment, improves roll rate and control, fixes stall at wing tips from high sweep for wings

Category	Design Feature	Reason
Airframe Layout	Lack of dihedral	Better survivability because allows pilot to fly upside down.
Airframe Layout	High wing sweep	Better transonic and supersonic performance
Materials	Zinc coating	For 25-year maritime environment durability
Materials	Aircraft grade aluminum, Polymer reenforce carbon fiber, and titanium	Reduces weight, strength in maritime environments
Propulsion	F404 Engine	Reduces production costs and minimized developmental risk
Propulsion	Afterburner	Better dash performance
Ordnance	Internal weapons bay	Reduced radar cross section
Carrier Suitability	Nose gear launch bar	Required to use the catapult
Carrier Suitability	Arresting hook	Essential for carrier recovery
Carrier Suitability	Reinforced main landing gear	Strengthened gear to withstand high-impact carrier landings and catapult loads
Survivability	Ejection seats	To save pilots incase of catastrophe

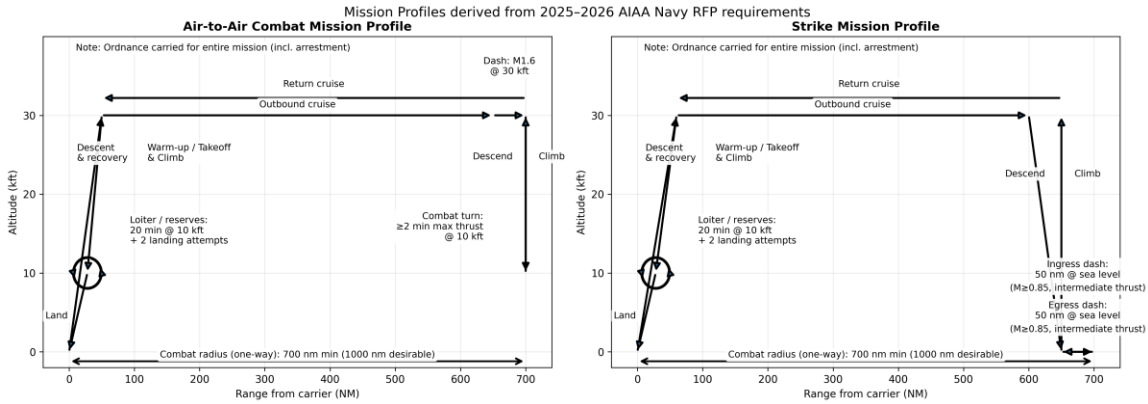


Figure 1-1. OV-1 and OV-2

1.2 Status of the Design

The current conceptual design is progressing through early-stage aerodynamic and configuration development and reflects a carrier-capable multirole fighter optimized for maneuverability, mission endurance, and system modularity. The layout has been shaped around the RFP's performance requirements, including the 700-nautical mile combat radius, dash capabilities, and carrier compatibility. While the baseline

configuration meets many of the mission constraints, several key assumptions and simplifications remain in this iteration that will be addressed in future refinement cycles. Landing gear loads have been estimated using Roskam's method to approximate sizing for both nose and main gear components. The gear locations are currently preliminary and are expected to shift as mass distribution and center-of-gravity (CG) tracking are finalized. For propulsion sizing, a takeoff parameter of 0.95 was assumed and used within Roskam's framework to determine required thrust for unassisted takeoff. This parameter will be reevaluated as more detailed constraint analyses are completed.
Several known deficiencies and modeling issues remain in the current design. Most notably, the vertical tail currently uses the same cambered airfoil as the wing, which introduces unwanted asymmetry and may affect yaw stability. This will be corrected by selecting a symmetric vertical airfoil in the next version. Additionally, the full range of takeoff conditions has not yet been explored. The team is currently comparing scenarios such as 90% of C_{L_max} versus a catapult-assisted launch with a 10-foot drop at the end of the deck to determine the limiting condition for thrust sizing. This analysis is critical to confirm takeoff field performance and sizing margins.
For aerodynamic performance analysis on area ruling and control effectiveness are still needed as they are critical for a fighter operating across a wide flight envelope. Further, the wing currently includes only a single flap, and lacks dedicated ailerons and flaperons. The next design phase will implement a multi-surface configuration to improve control authority during both high-speed maneuvering and low-speed carrier recovery. This change will also require updated calculations for exposed and wetted wing area.
Pitch control in the current design is based on a fixed horizontal stabilizer with an elevator. To enhance maneuverability and pitch responsiveness, this will be replaced with an all-flying tail (stabilator) in the next iteration. The CG location, influenced by a wing-fuel layout, is currently positioned aft of the aerodynamic center. While this configuration is theoretically manageable using a fly-by-wire control system, it may reduce static margin and limit inherent longitudinal stability. A forward shift in the wing root's leading-edge intercept or redistribution of internal mass may be needed to improve stability characteristics. Further investigation will determine whether such changes are structurally and operationally beneficial, or whether reliance on active control is sufficient given the aircraft's mission profile and technology level.
The inboard profile of the aircraft is still under development and has not yet been finalized. This component of the design defines the internal arrangement of major

structural elements, fuel tanks, avionics bays, landing gear wells, and systems routing. Its completion is dependent on further refinement of the external geometry and sizing of internal components. While preliminary layouts have been explored, a full inboard profile will be developed in an upcoming design phase as subsystem packaging and structural volumes are better defined.

Flight loading conditions for various mission segments have not yet been evaluated. This includes maneuver, gust, and landing loads critical for structural sizing and compliance with safety margins. These loads will be determined in the next semester as part of AE460B and will be based on the aircraft's expected mission envelope, aerodynamic behavior, and structural layout. The absence of this analysis currently limits the accuracy of structural weight estimates and must be completed to support airframe stress evaluations and material selection.

Structural analysis of the airframe is also planned for next semester and will follow the identification of load cases and material assignments. This will include modeling the wing box, fuselage frames, and load paths under critical conditions such as high-g maneuvers, landing impact, and arrested recovery. Without this analysis, the design currently assumes generalized structural sizing factors, and as such, detailed stress validation and weight distribution remain incomplete.

The next iteration of the design will incorporate the above updates to improve aerodynamic fidelity, control surface effectiveness, and static stability. Each design modification will be fully documented in Section 2 of this report, including supporting rationale and any resulting impacts on sizing, weight, and mission performance.

2 System Requirements

2.1 Identification and Allocation

SRD Paragraph Number	Requirement (full)	Aerodynamic	Performance	Airframe/Subsystems	Stability & Control	Structures	Consultant	Required	Desired	Achieved	Legacy Aircraft (FA-18E/F)
3.1.a	The aircraft may include any combination of survivability, vulnerability and effectiveness features to maximize its cost-effective capability.	x	x	x	x	x	X				
3.1.b.a	Aircraft may be single or multi-engine			x							
3.1.b.b	Engines must be existing production design.			x							
3.1.c.a	Materials suitable for 25-year service life in maritime environment.	x	x	x	x	x	x	25 years		In Progress	
3.1.c.b	Materials must be available in production quantities.	x	x	x	x	x	x				
3.1.d	Subsystems (avionics/mission systems) at TRL 6 or higher.			x				TRL 6+			
3.1.e.a	External carriage of ordnance acceptable.			x						All Internal	
3.1.e.b	Internal carriage desirable for reduced RCS.			x							
3.2.a	Operable from CVN-68 and CVN-78 class carriers.		x				x				
3.2.b	Minimal impact on carrier air wing operations.		x				x				
3.2.1.a	Capable of launch on tropical day (89.8°F) with		x				x				

SRD Paragraph Number	Requirement (full)	Aerodynamic	Performance	Airframe/Subsystems	Stability & Control	Structures	Consultant	Required	Desired	Achieved	Legacy Aircraft (FA-18E/F)
	zero WOD.										
3.2.1.b	Catapult minimum end airspeed requirement highest of the following conditions: i. An end speed which results in the CG position of the aircraft sinking no more than 10 feet from its position at the end of the power stroke, with a deck run not to exceed 32 feet (distance from the end of the power stroke to round-down), with cockpit control position held either fixed or free or controls active. ii. The speed represented by 90% of the maximum lift coefficient, power off, out of ground effect. iii. The minimum airspeed at which the aircraft has a longitudinal acceleration of 0.065 g (2.0913 ft/sec2) at zero flight path angle. iv. If multi-engine, the minimum aircraft control speed with one engine inoperative						x			183 knots	135 knots
3.2.2.a	Arrestment engaging speed is assumed to be 5% above approach speed.		x								
3.2.2.b	10% above stall < Approach airspeed < 145 kt.		x							144 knots	

SRD Paragraph Number	Requirement (full)	Aerodynamic	Performance	Airframe/Subsystems	Stability & Control	Structures	Consultant	Required	Desired	Achieved	Legacy Aircraft (FA-18E/F)
3.2.2.c	Arrestment landing weight shall include sufficient fuel for 20 minutes loiter at 10,000 ft and two landing attempts, 25% maximum fuel weight, and 50% store weight.		x								
3.3.1.a	Avionics weight 2,500 lb.	x	x	x	x	x	x	2500 lbs	N/A		
3.3.1.b	Air-to-air ordnance load shall be: i. AIM-120C QTY 6. ii. AIM-9X: QTY 2	x	x	x	x	x	x				
3.3.2.b	Strike ordnance load shall be: i. MK-83 JDAM QTY 4. ii. AIM-9X: QTY 2	x	x	x	x	x	x				
3.4.1.a	700 nm minimum combat radius							700 nmi	1000 nmi	800 nmi	400-600 nmi
3.4.1.b	2 minutes combat at max thrust at 10,000 ft.							2 min	5 min	In Progress	
3.4.1.c	Ordnance carried for entire mission	x	x	x	x	x	x				
3.4.1.d	8 deg/sec sustained turn rate at 20,000 ft. at mid mission weight							8° /sec	10° /sec		~8-9 °/sec
3.4.1.e	Mach 1.6 dash at 30,000 ft.	x						1.6	2.0	1.49	1.6
3.4.2.b	50 nm sea-level dash segments, intermediate thrust.									Include in range	
3.4.2.c	Sea-level dash speed Mach ≥0.85.							< .85	< .90	0.81	~0.85
3.5.a	Vertical load factor >7g mid-mission weight	x	x	x	x	x	x	7g	8g		7.5 g
3.5.b	External store carriage ≥10,000 lb total or ≥3,000	x	x	x	x	x	x	10,000 lb total;			17,750 lbs total

SRD Paragraph Number	Requirement (full)	Aerodynamic	Performance	Airframe/Subsystems	Stability & Control	Structures	Consultant	Required	Desired	Achieved	Legacy Aircraft (FA-18E/F)
	lb per store.							3000 lb /store			
3.5.c	Launch conditions must be met with C-13-2 catapult.		x				x				
3.5.d	Recovery requirements must be achievable with a WOD not greater than 15 knots and using the Mark 7 Mod 3 arresting gear		x				x				
3.5.e	SEROC (single engine rate of climb) ≥200 ft/min (launch, multi-engine).							≥200 ft/min	N/A		~200-300 ft/sec
3.5.f	Approach SEROC ≥500 ft/min.							≥500 ft/min	N/A		
3.5.g	Unfolded wingspan ≤60 ft.							≤60 ft	N/A	41.5 ft	44.9 ft
3.5.h	Folded wingspan ≤35 ft.							≤35 ft	N/A	37.5 ft	30.7 ft
3.5.i	Aircraft length ≤50 ft.							≤50 ft	N/A	50 ft	60.3 ft
3.5.j	Aircraft height ≤18.5 ft.							≤18.5 ft	N/A	14.16 ft	16.0 ft
3.5.k	Spot factor minimized (folded footprint).										
3.5.l	Maximum take-off gross weight ≤90,000 lb.							90,000 lb	N/A	42,567 lb	66,000 lb

2.2 Design Changes Required

The following requirements are currently not fully satisfied and have been designated Yellow or Red based on the project's current design status. Each is expected to be addressed in subsequent iterations of the aircraft design, with the intent of achieving full compliance. Supporting rationale and planned design adjustments are provided for each item.

The requirement for materials capable of withstanding a 25-year service life in a maritime environment is currently rated Yellow. While a general material plan has been outlined, comprising aircraft-grade aluminum, titanium alloys, and polymer-reinforced

carbon fiber. Specific materials have not yet been assigned to individual components. The team intends to implement corrosion-resistant surface coatings on metallic elements and select composite materials that are proven for durability in salt-fog and high-humidity environments. Once material distribution and surface protection strategies are fully defined, this requirement is expected to be Green.

The aircraft currently features an internal weapons bay configuration only, which improves low observability but limits flexibility in ordnance deployment. This poses an issue for missile systems such as the AIM-9X and AIM-120, which often require lock-on prior to launch. To resolve this, the design will be updated to include external hardpoints, enabling carriage of air-to-air missiles in a semi-recessed or wing-mounted configuration. This modification allows for higher engagement accuracy while preserving stealth when needed. The configuration will include modular pylons or dual-mode attachment points to maintain adaptability.

The catapult minimum end airspeed requirement is currently being evaluated against four possible criteria, with the highest value required to be met. At present, the calculated end speed is 183 knots, which is within an acceptable range but remains on the higher side since the legacy aircraft is capable of 135 knots. The design team is actively comparing the various conditions, including 90% of maximum lift coefficient and a 10-foot sink from catapult end speed, to determine the governing case. Future iterations will adjust wing loading and high-lift device sizing to reduce required end speed and ensure compliance.

The two-minute combat maneuvering segment at 10,000 ft under maximum thrust is feasible; however, the impact on fuel consumption and overall mission radius has not yet been fully quantified. Although the aircraft can theoretically perform the segment, there is concern that it may reduce available fuel for outbound or return cruise segments. A full mission fuel breakdown is planned to assess and mitigate this tradeoff, either through increased fuel fraction or engine efficiency improvements.

The sustained turn rate requirement of 8 degrees per second at 20,000 ft and mid-mission weight is also currently rated Yellow. Preliminary estimates suggest the aircraft may fall slightly short of this threshold due to drag levels and available thrust at altitude. The team is actively refining drag polar inputs, evaluating sustained turn models, and assessing potential aerodynamic optimizations such as revised control surface sizing and slight reductions in wing loading. These changes are expected to yield the required

turn performance once implemented and validated against mid-mission weight conditions.

The vertical load factor (Nz) at mid-mission weight has not been evaluated as it is contingent on both aerodynamic limits and structural design. If the aircraft runs out of lift at high-g conditions before reaching structural limits, it will be aerodynamically constrained; otherwise, the structural limit load factor will govern. Structural analysis is scheduled for the following semester, at which point a definitive Nz capability will be calculated and incorporated into the design loop.

Both the single-engine rate of climb (SEROC) requirements, for launch (≥200 ft/min) and approach (≥500 ft/min), are still undergoing calculation. These metrics are sensitive to engine thrust, aircraft weight, and drag configuration. They will be addressed as part of the ongoing propulsion and climb performance analyses, and updated engine selections are expected to help satisfy these thresholds.

For the dash speed requirement of Mach 1.6 at 30,000 ft, the aircraft currently achieves only Mach 1.49. This shortfall is primarily attributed to excessive drag, stemming from a high wing surface area. Planned changes include reducing wing area and switching to a more powerful engine. The F404 engine used in earlier sizing iterations was chosen to manage cost, but with the project well under budget, a more capable engine is now feasible and justified to close this performance gap.

At sea level, the aircraft currently achieves a dash speed of only Mach 0.81, which is below the expected performance level. To address this, the team is evaluating aerodynamic refinements such as inlet shaping, area ruling, and potential reductions in parasitic drag. Propulsion upgrades under consideration for high-altitude dash will also improve low-altitude thrust margin.

The aircraft is expected to complete the 50 nm sea-level dash segment in theory, but the range impact of this high-drag, high-fuel-consumption segment has not yet been quantified. The mission fuel model will be expanded to incorporate this requirement and validate that the aircraft retains sufficient fuel for return cruise and recovery following the dash.

The external store carriage requirement, supporting at least 10,000 lb total or 3,000 lb per station, has not yet been structurally verified as they fall under the broader structural analysis timeline. Based on initial sizing, this requirement is expected to be met once detailed structural modeling is complete.

The aircraft currently has a folded wingspan slightly above the 35-foot limit due to oversized flaps. With only two feet of folding span available per side, the current folded configuration does not meet the required limit. Planned flap redesign and optimization of hinge placement will allow additional span to be folded, bringing the wingspan under the limit. This is also linked to the aircraft's spot factor, which is close to but not yet fully compliant. The design team is prioritizing geometric refinements to meet both folded span and spot factor constraints simultaneously.

Each of these Yellow-rated requirements has a defined path to resolution and will be addressed in the next phase of design through aerodynamic, structural, or propulsion-related updates. With these planned modifications, the design is expected to achieve full compliance with the RFP requirements in future iterations.

3 Aircraft Description

3.1.1 Preliminary Weight Estimate

A preliminary weight estimate is performed for a "jumping off point" to be achieved. By utilizing Roskam's Part I as well as key assumptions being made, a weight estimate for both the air to air and the strike mission are produced. To ground this plane as well as a sanity check, these weights are cross referenced with the F-18, as it is a carrier equipped strike fighter jet with a similar goal profile that we will achieve. These weights are shown below.

Preliminary Weight: Strike			Weight (lb)
Operating Empty Weight (W_OE)			26236.179
	Empty Weight (W_E)		25876.179
	Trapped Fuel and Oil (W_tfo)		120
	Crew Weight (W_CREW)		240
Fuel (W_F)			18276.179
W_OE + 60% Fuel			37201.886
W_OE + W_F			44512.357
Payload Weight (W_PL)			7600
W_OE + WF + W_PL			52112.357

Table 3-1. Weight estimate of F-44 Falcon, strike mission

Preliminary Weight: A2A			Weight (lb)
Operating Empty Weight (W_OE)			25167.589
	Empty Weight (W_E)		24807.589
	Trapped Fuel and Oil (W_tfo)		120
	Crew Weight (W_CREW)		240
Fuel (W_F)			19307.589
W_OE + 60% Fuel			36752.142
W_OE + W_F			44475.177
Payload Weight (W_PL)			5500
W_OE + WF + W_PL			49975.177

Table 3-2. Weight estimate of F-44 Falcon, A2A mission

After cross referencing this data against known weights of similar aircraft such as the F-18, the weights are not only reasonable, but a fantastic ballpark range to begin designing from. It should be noted that the weight difference between each mission is directly related to the required range, which impacts the fuel, and the types of explosive stores required to complete the mission. The overall design of the aircraft will be able to execute each mission independently. For these calculations to be made, a few key assumptions must be clarified. For starters, the trapped fuel/oil has not been derived, but we understand it is not an extreme weight factor. So, to be conservative, we are working with the assumption that the total trapped/unusable fuel and oil to be approximately 120 lbs. The other two key assumptions were the empty weight fraction of 0.5 and a fuel allowance factor of 0.06.

The excel calculations are shown below.

Inputs (Strike)	
Payload W_PL (lb)	7600
Empty-weight fraction ε (decimal)	0.5
Fuel allowance δ (decimal)	0.06
Outputs (Strike)	
Π (product of factors)	0.706853215
Raw fuel fraction (1−Π)	0.293146785
Final ϕ (add δ)	0.353146785
W_0 (lb)	51752.35702
$W_e = \varepsilon \cdot W_0$ (lb)	25876.17851
$W_f = \phi \cdot W_0$ (lb)	18276.17851

Table 3-3. Excel weight calculations including assumptions and fractions for the strike mission.

A2A Mission (700 nm out & back, 2-min AB, 30-min reserve)	
Inputs (A2A)	
Payload W_PL (lb)	5500
Empty-weight fraction ε (decimal)	0.5
Fuel allowance δ (decimal)	0.06
Outputs (A2A)	
Π (product of factors)	**0.670853176**
Raw fuel fraction (1−Π)	**0.329146824**
Final ϕ (add δ)	**0.389146824**
W_0 (lb)	**49615.17725**
$W_e = \varepsilon \cdot W_0$ (lb)	**24807.58862**
$W_f = \phi \cdot W_0$ (lb)	**19307.58862**

Table 3-4. Excel weight calculations including assumptions and fractions for the A2A mission.

3.1.2 Wing Area (S) and Takeoff Thrust (T_TO) Estimate (Roskam, Part I)

Every airplane that intends to fly requires wings and thrust, both of which can become convoluted and complicated very quickly. Because of this, using resources such as Roskam's Part I is an essential part of the preliminary design process. Roskam outlines chapter 3, section 2.5.2 says that for carrier based, planes there is a specific methodology that can be used to obtain wing sizing. This is shown below.

$$W_{TO} = \frac{\left(0.5\rho(V_{WoD} + V_{cat})^2 S C L_{MAX_{TO}}\right)}{1.21}$$

Equation 3-1

Because we have already found an estimated take-off weight, a plot of wing loading versus thrust to weight ratio can be manufactured to understand if the chosen wing area allows for a feasible aircraft.

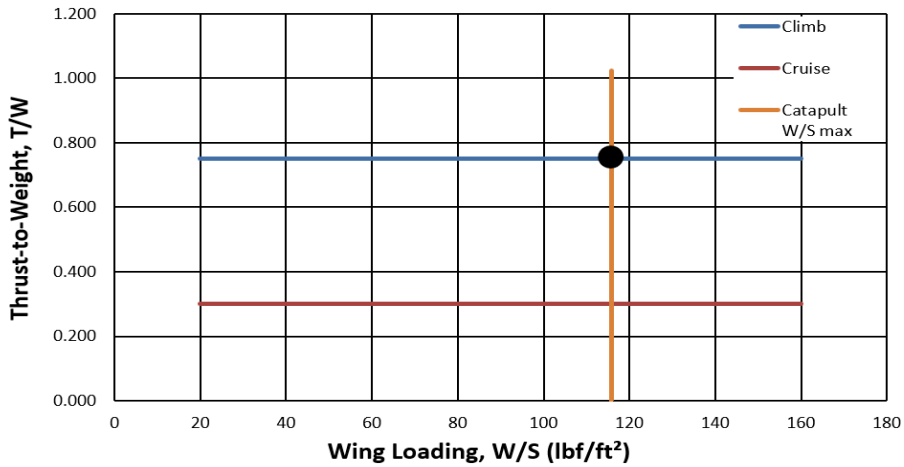


Figure 3-1. Wing Loading vs. Thrust to Weight

	Assumptions	
	Unit	Value
S	ft^s	400
CL_Max, TO	-	1.6
V_WoD	ft/s	0
TOP	-	0.95
W_TO	-	43000

Figure 3-2. Assumptions used to find takeoff thrust

Wing Area

As a baseline, several initial assumptions are required. The F/A-18 Hornet serves as a useful reference due to its proven performance and efficient wing design, so an initial wing area similar in magnitude was selected as a starting point. This value is intended only for preliminary sizing and will be adjusted later to account for range and endurance requirements. The current configuration includes a larger wing area, which results in a lower wing loading and easily satisfies the carrier takeoff constraints, but introduces trade-offs in other areas of the design. For the purposes of this preliminary analysis, a slimmer wing area is therefore used to better represent a realistic design target and to provide a more meaningful assessment of the wing loading and thrust requirements.

Coefficients of Lift

Using Roskam's Part I, there is a given range of take-off lift coefficients for a strike fighter aircraft. This has led us to choose a middle ground value to keep our calculations conservative while giving us room to create a plane that is still theoretically possible.

For the cruise and climb coefficients of lift, we again chose middle ground values as a base assumption to get an idea of what thrust to weight ratio we would need.

Wind Over Deck

This is a specific parameter outlined in the SRD given by AIAA. It specifies that this aircraft must be able to take off with catapult assistance without any additional WOD velocity.

Take Off Parameter

A takeoff parameter of TOP = 0.95 was chosen using Roskam Part I for guidance for military jet aircraft operating from aircraft carriers. Because carrier aircraft rely on catapult launch rather than runway ground roll, lower TOP values are appropriate. The selected value reflects tropical day, zero wind over deck conditions specified in the SRD and provides reasonable margin while remaining consistent with the performance of existing carrier-based strike aircraft.

Take-Off Thrust

The design point is shown in Figure 3-1 by the black marker at the intersection of the selected thrust-to-weight ratio and the allowable wing loading. Using the assumptions summarized in Figure 3-2, the maximum permissible wing loading is determined by carrier catapult capability. Its intersection with the assumed maximum thrust-to-weight ratio defines the design point, which can then be used to back-calculate the total thrust required from the propulsion system to satisfy carrier takeoff and climb requirements.

Take-Off Weight

Using Roskam's estimations initially provided a value shown in Table 3-1 of a maximum of roughly 52000 lbs.

Catapult Considerations

As outlined in the SRD, a C13-2 deck catapult will be used to assist the F-44 in reaching the required take-off velocity. Using this chart provided by Professor Butler, we can see that the added velocity due to the catapult is roughly 165 knots. This is the value used in equation 3-1 after being converted into appropriate units. The chart provided to us is shown below in Figure 3-3.

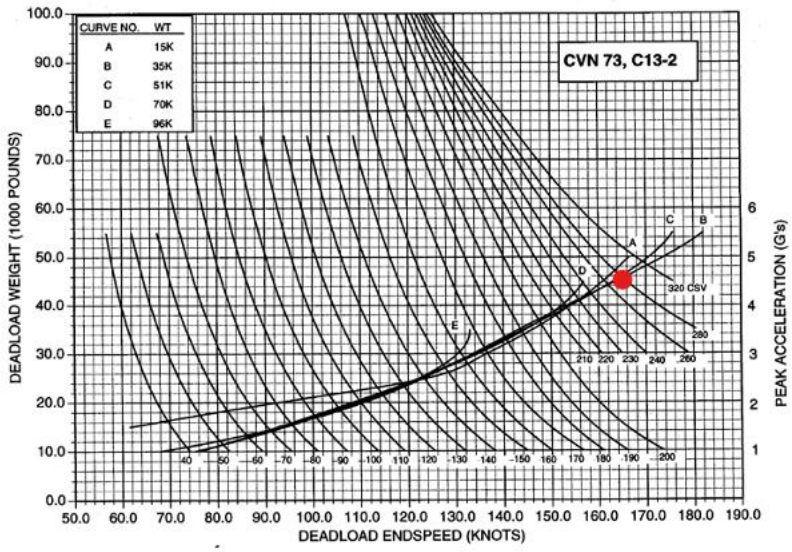


Figure 3-3. Deadload End speed vs. Deadload Weight for the C13-2 carrier catapult.

3.1.3 Landing Gear Loads and Tire Sizing (Roskam, Part IV)

Landing Gear Loads and Tire Sizing

Preliminary landing gear loads and location were determined using previously obtained weight estimates derived with formulas found in “Roskam’s Airplane Design Part IV: Layout of Landing Gear and Systems.” An estimate for the loads that the nose and main gears will encounter were predicted. These loads were calculated as a percentage of the maximum ramp weight (1.01 times the max takeoff weight), with the total nose gear load being ten percent (10%) of the maximum ramp weight and the total main gear load being ninety percent (90%) of the maximum ramp weight. The main and nose landing gear are positioned 180 inches from each other, with each main gear positioned equidistant from the centerline of the aircraft as shown in Figure 3-4.

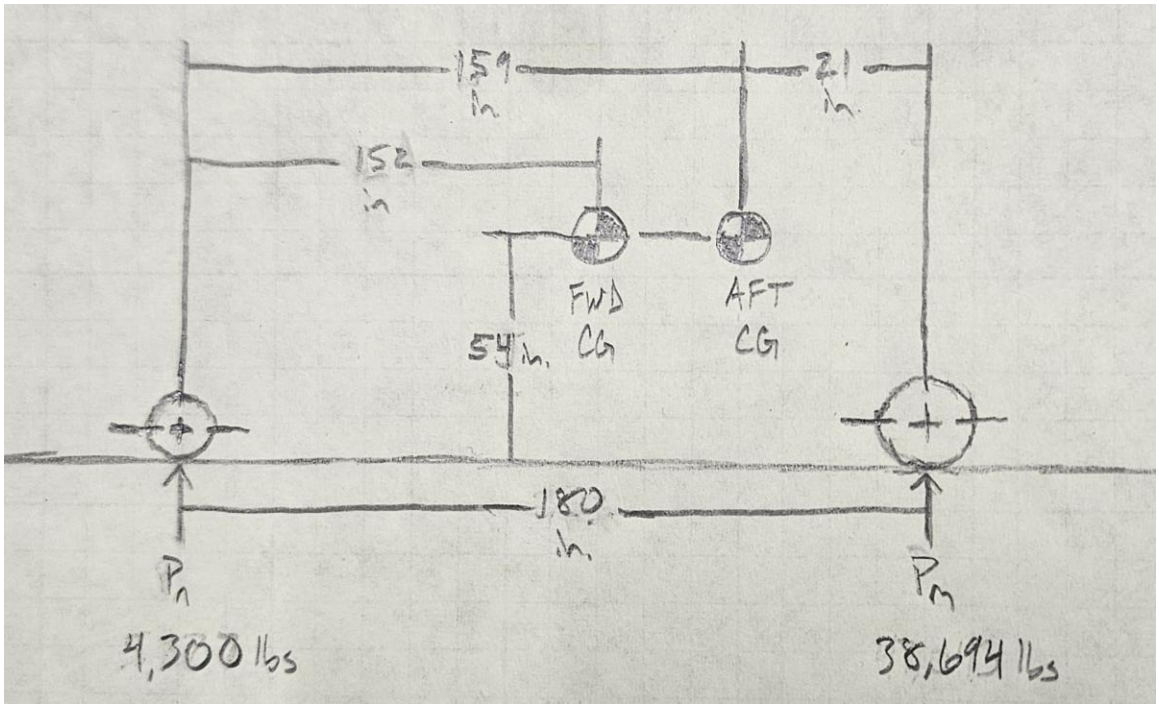


Figure 3-4. Landing gear layout sketch with expected loads for nose and main gear

Tire Selection

The static load on the nose gear was found to be 4,300 lbs and the static load on the main gear was found to be 38,694 lbs. Each tire on the nose and main gear need to have a static load rating of at least half of the total static load on each respective strut. The nose gear will have two tires to distribute the load on, while the main gear will have two struts with one tire on each to distribute the load. The tires chosen were 27.75x8.75-14.5 24-ply for the main gear and 14x5.5-6 8-ply for the nose gear, as shown in Table 3-5 below.

Table 3-5. Tire Specifications

	Size	**Ply Rating**	**Rated Speed (knots)**	**Rated Load (Lbs)**	**Max Braking Load (Lbs)**	**Max Bottoming Load (Lbs)**	**Rated Inflation (psi)**	**Max Diameter (in)**
Main	35x11.5-17	26	208	28,300	42,450	76,460	255	35
	32x11.5-15	26	210	27,800	41,700	83,400	290	32
	30x11.5-14.5	26	220	26,600	38,570	71,800	265	29.75
	27.75x8.75-14.5	24	225	21,500	31,175	58,050	320	27.75
Nose	23x7-12	12	182	7,800	11,700	23,400	160	23.2
	22x6.6-10	22	195	14,150	21,225	42,450	304	22.2

	Size	Ply Rating	Rated Speed (knots)	Rated Load (Lbs)	Max Braking Load (Lbs)	Max Bottoming Load (Lbs)	Rated Inflation (psi)	Max Diameter (in)
	21x7.25-10	20	195	12,000	18,000	36,000	325	21.25
	18x5.7-8	20	217	9,000	13,500	27,000	315	17.9
	17.5x5.75-8	12	182	5,000	7,500	15,000	180	17.5
	14.5x5.5-6	8	186	2,890	4,335	8,670	123	14.2

Discussion of Tire Specifications

The selected main gear tire size 27.75x8.75-14.5, 24-ply is designed to support a static load of 21,500 lbs per tire, ensuring adequate capacity for high loads experienced during takeoff, landing, and ground operations. The 320-psi inflation pressure and 225 knots rating make them suitable for high-speed takeoff and landings with extended taxiing. The max diameter of the tire of 27.75 inches allows for a compact fitment within the fuselage of the aircraft.

The nose landing gear uses 18x5.7-8, 20-ply tires, appropriate for the lower loads on the nose gear, with a static load capacity of 9,000 lbs per tire. A maximum inflation pressure of 315-psi provides sufficient support for steering and braking forces. The tires also have a 217 knots speed rating which is sufficient for takeoff and landing speeds and very similar to the main landing gear tire speed rating (225 knots). Even though there are smaller tires that meet the static load requirement, the speed ratings for those tires are not sufficient enough to handle the aircraft's takeoff speed. The maximum tire diameter of 17.9 inches enables an efficient integration within the aircraft fuselage.

Overall, these tire selections exceed aircraft performance, safety, and reliability requirements to allow for a safety factor while being the smallest tire choices available. With a maximum takeoff speed of 183 knots and an approach speed of no more than 140 knots, the combination of high load capacity, appropriate inflation pressures, and speed ratings ensures safe takeoff, landing, and taxi operations.

3.2.1 Design Evolution

Figure 3-5. F/A-18E/F Super Hornet

The design evolution of our aircraft is as follows from Figure 3-6 to Figure 3-9. The preliminary designs were based on initial research of current carrier-based fighter aircraft. The sketched aircraft reflects the integration of research and preliminary analysis. The United States Navy's current carrier-based strike fighter aircraft, the F/A-18E/F Super Hornet played a major role in inspiring our design. The F/A-18E/F Super Hornet seen in Figure 3-5 has many distinct characteristics that are shared with our aircraft design such as dual engine intake, dual engine configurations, dual canted vertical stabilizers, and horizontal stabilizers below the wing centerline. These characteristics are what enabled us to design a next generation carrier-based strike fighter.

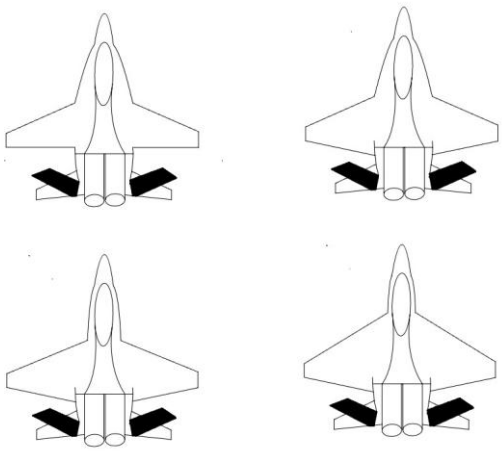


Figure 3-6. Preliminary Aircraft Wing Configurations

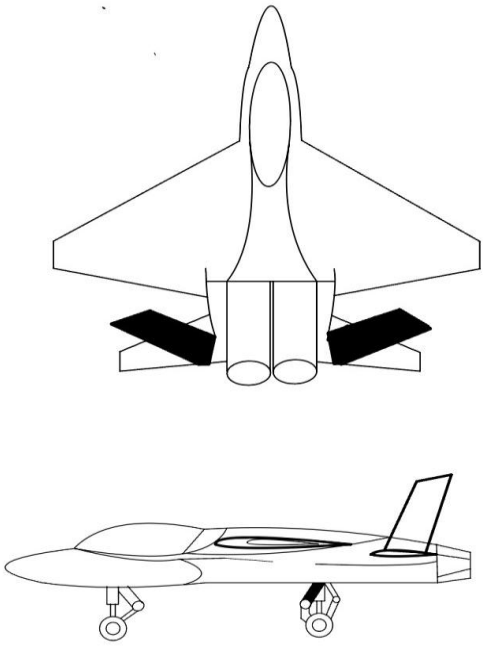


Figure 3-7. Preliminary Aircraft Sketch

Figure 3-8. Preliminary CAD Model

Our preliminary design was based on an estimated initial aircraft weight of 50,000 lbs. This was calculated based on the mission parameters for a strike and air-to-air mission. Our design focused on range, cruising speed, and dash speed at sea level and 30,000 ft altitude. Our team developed a wing layout excel calculator to graph the leading edge and trailing edge of our main wing, horizontal stabilizers, and empennages to find the best suiting wing structure for our aircraft. The driving parameters for this process are the wing area, span, swept angle, taper ratio, and the nose leading edge root chord distance. Through this process we were able to use SolidWorks to 3D model our preliminary design. This approach allowed us to visualize our aircraft and eventually improve mission effectiveness.

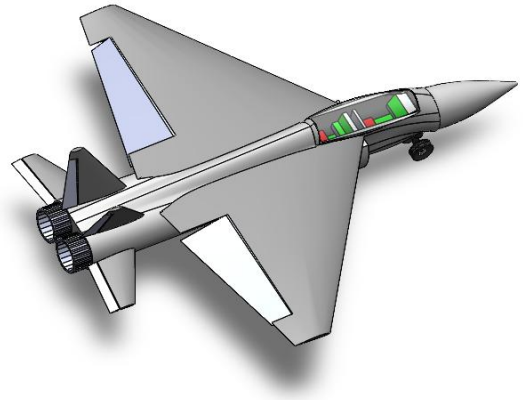


Figure 3-9. Final CAD Model

The final CAD model progressed with the same dual intake, dual engine, and dual vertical stabilizer layout. Changes were made to the wing placement and sizing as well as fuselage length to height ratio. The final CAD model had multiple additions such as the cockpit with seats for a pilot and a co-pilot, high lift devices, and an internal payload system.

3.2.2 Design Requirements on the Configuration

3.2.2.1 Cockpit Location

The cockpit is positioned at the front of the aircraft to provide the pilots with a clear, forward, and downward view. This is critical for pilots to maintain situational awareness during a mission, but more specifically takeoff, landing, and taxi operations on the aircraft carrier. This forward positioning allows the pilots to effectively control the aircraft and monitor onboard systems from a forward point of view. Placing the cockpit just aft of the nose helps avoid aerodynamic disturbances at subsonic and supersonic speeds. This allows for efficient airflow along the fuselage and into the leading edges of the wings. Lastly, the forward cockpit helps with the center of gravity combined with the aircraft's structural layout.

3.2.2.2 Airfoil Choices

For the airfoil selection, a survey was conducted of airfoils with well-documented subsonic and supersonic performance data provided by the National Advisory Committee for Aeronautics (NACA). The lift and drag characteristics of three candidate airfoils were compared, leading to the selection of the NACA 65-206 airfoil (Figure 3-10 & Table 3-6).

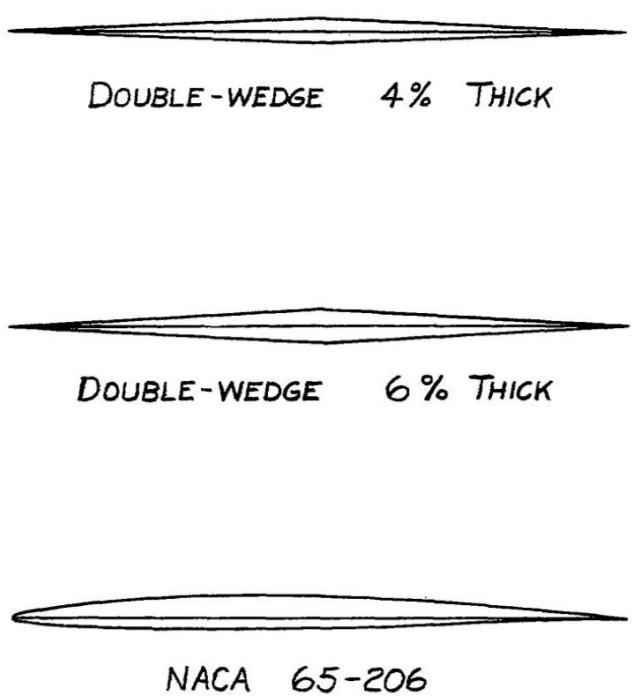


Figure 3-10. Comparison of Airfoil Profiles. Reproduced from Solomon and Henney (1947).

Although the blunt leading edge may appear unfavorable at supersonic speed, particularly when compared to the sharper leading edges of Double-Wedge profiles, it was determined that carrier-based performance is more critical. The chosen airfoil offers better low-speed handling and efficiency, which are essential for safe and effective carrier-based landings, outweighing its supersonic disadvantages.

When calculating airfoil performance, the necessary data was taken directly from Roskam's reference, as shown the Table 3-7.

Table 3-6. NACA 65-206 experimental data

Description	**Variable**	**Value**	**Units**
Airfoil zero lift angle of attack	α_{0_l}	-1.6	degrees
Zero lift coefficient moment	c_{m_0}	-0.031	1/rads
Airfoil lift curve slope	c_{l_α}	0.105	1/degrees
Aerodynamic center	$a.c.$	$0.257\overline{c}$	rads
Airfoil angle of attack at maximum lift	$\alpha_{c_{l_{max}}}$	12	degrees
Airfoil maximum lift coefficient	$c_{l_{max}}$	1.03	rads
Airfoil linear range of angle of attack	$\alpha^{\cdot}$	6	degrees

3.2.2.3 Wing Placement

The wing placement features a high-mounted configuration integrated with the fuselage with a carry-through system. The nose to root leading edge length is 250 inches which was optimized for the aircrafts center of gravity position. We opted for high-mounted configuration to have an internal payload system directly below the wing configuration. The high wing configuration helps with short field performance by generating stronger wing to fuselage lift interactions, giving higher lift at lower speeds. This is to enhance takeoff and landing performance. Additionally, the high wing is above the aircraft's center of gravity which in turn provides a pendulum-like stability effect. This improves roll

behavior and stall behavior at high angles of attack. Lastly, keeping the wing above the fuselage centerline helps reduce interference drag and prevent boundary layer ingestion from the fuselage, essentially helps maintain clean airflow over the wing.

Table 3-7. Wing Derived Geometry

Aspect Ratio AR	2.870416667
Semi-span (in)	249
Root chord c_r (ft)	24.09638554
Root chord c_r (in)	289.1566265
Tip chord c_t (in)	45.3372
MAC (in)	199.1967871
y_bar (in)	96.83333333
m_LE = tan(Λ_LE)	0.932515086
LEMAC X (in) = XLE(y_bar)	340.2985442
TEMAC X (in)	539.4953313
Tip chord c_t (ft)	3.7781

3.2.2.4 Planform, Sweep, Taper Ratio, Aspect Ratio, Dihedral/Anhedral

Planform Area: 600 ft^2, large wing surface area to generate lift for carrier takeoff as well as fuel consuming at altitude. This larger surface area was chosen to improve low-speed performance during takeoff, landing, and climbing. Additionally, the larger wing was chosen to provide more internal fuel volume and structural stability. A large wing distributes aerodynamic loads more effectively, which in turn improves maneuverability.

Sweep Angle: A high sweep angle of 43 degrees provides substantial aerodynamic benefits such as delaying shock formation and reducing wave drag when the aircraft goes from transonic to supersonic speeds. This allows the aircraft to maintain higher Mach numbers with increased efficiency. This sweep angle helps preserve lift and stability during high-speed maneuvers while also managing the shit of center of pressure when nearing Mach 1.

Taper Ratio: 0.2, this taper ratio means out tip chord is 20 percent of the root chord. This smaller taper ratio improves aerodynamic efficiency by creating a lift distribution and reducing the induced drag. This taper ratio also approaches an elliptical shape which is an optimal wing shape. This taper ratio enhances performance during high lift conditions, improving maneuverability. Lastly, the taper reduces the weight of the wing by lowering the structural loading near and at the wingtips.

Aspect ratio: 2.87, which is a relatively low aspect ratio. This is to improve maneuverability and ensure the structural integrity is sound. The shortened span allows for high-G maneuvers and allows for improved roll performance by reducing the roll inertia. A smaller aspect ratio also promotes more stable vortex formation near the leading edge, which in turn helps maintain lift during aggressive maneuvers such as high-G's during fast turn rates.

3.2.2.5 Tail Sizing and Tail Placement

The horizontal and vertical stabilizers are positioned aft of the aircraft to ensure aircraft stability and control. This is to provide the necessary moment of arm for effective stability and control. Positioning these surfaces at the rear increases their leverage relative to the center of gravity, which in turn improves pitch and yaw authority. This also allows for minimizing the size of the stabilizers while maintaining the same control effectiveness. The horizontal stabilizer has a surface area of 90 ft^2 with an aspect ratio of 4.01 and a reference span of 19 ft. The higher aspect ratio allows for improved longitudinal stability and cruise efficiency. The vertical stabilizers have a surface area of 50.88 ft^2 per vertical stabilizer with an aspect ratio of 2.83 which is like our main wing. The vertical stabilizers have a semi span of 6 ft. This aspect ratio and surface area help maintain effective yaw control, especially at high angles of attack.

Table 3-8: Horizontal Stabilizer Derived Geometry

Aspect Ratio AR_H	4.011111111
Semi-span (in)	114
Root chord c_r,H (ft)	6.766917293
Root chord c_r,H (in)	81.20300752
Tip chord c_t,H (in)	32.48120301
MAC_H (in)	60.32223416
ȳ_c,H (in)	48.85714286
m_LE,H = tan(Λ_LE,H)	0.554309051
LEMAC_H X (in) = XLE_H(ȳ_c,H)	45.78495651
TEMAC_H X (in)	106.1071907
Vh (Hor Tail Volume)	0.163376129

Table 3-9: Vertical Stabilizer Derived Geometry

Per-Fin Area S_V (ft^2)	50.885
Aspect Ratio AR_V	2.829910583
Semi-height (in)	72
Root chord c_r,V (ft)	7.067361111
Root chord c_r,V (in)	84.80833333
Tip chord c_t,V (in)	16.96166667
MAC_V (in)	58.42351852
z_bar,V (in)	28
m_LE,V = tan(Λ_LE,V)	0.75355405
LEMAC_V X (in) = XLE_V(z_bar)	521.0995134
TEMAC_V X (in)	579.5230319
Vv (Vert Tail Volume)	0.080271724

3.2.2.6 & 3.2.2.7 Control Surfaces & High Lift Devices

Our aircraft utilized three specific high lift devices for the control surfaces: flaps, elevators, and ailerons. These high lift devices are key components of aircraft to maintain stability, performance, and maneuverability.

Flaps: The flaps which are located on the trailing edge of wing, which extend during takeoff and landing. The flaps are used to increase the effective wing surface area and improve the aerodynamic performance for low-speed lift at takeoff and landing. Each flap per wing spans 80% of the wings span and covers 25% of the chord length. The flaps cover 80% of the span, leaving 10% span from the tip chord and 10% span from the root chord. This design is effective, but for future design the wing will have a flaperon and an aileron to maximize aerodynamic effectiveness.

Elevators: The elevators use 100% of the horizontal stabilizer span. This full span configuration maximizes control authority by allowing the entire trailing edge if the horizontal stabilizer contributes to pitch maneuvering. Each elevator has a surface area of 12.49 ft^2 which is sufficient with uniform aerodynamic loading for rapid pitch control. These parameters allow the aircraft to meet both the trim and maneuvering requirements.

Rudders: The rudders span 86.11% of the vertical stabilizer span with a surface area of 16.08 ft^2. These parameters ensure an effective control surface for managing yaw stability and directional control. This is essential for turns, weapon release, asymmetrical thrust, and cross wind operations. The vertical stabilizers are also canted, which means the rudders have the same canting configurations to avoid downwash from the canopy.

3.2.2.8 Payload Placement

The payload is located inside the fuselage aft of the cockpit and just below the wing carry through box. This payload placement ensures minimal aerodynamic interference by significantly reducing the drag. The only drag accounted for with this system is when the payload doors are open to drop bombs or fire missiles. The internal payload with closed doors when cruising allows for increased range and speed for each mission. The internal payload system optimizes the center of gravity and weight distribution, which enhances aircraft performance.

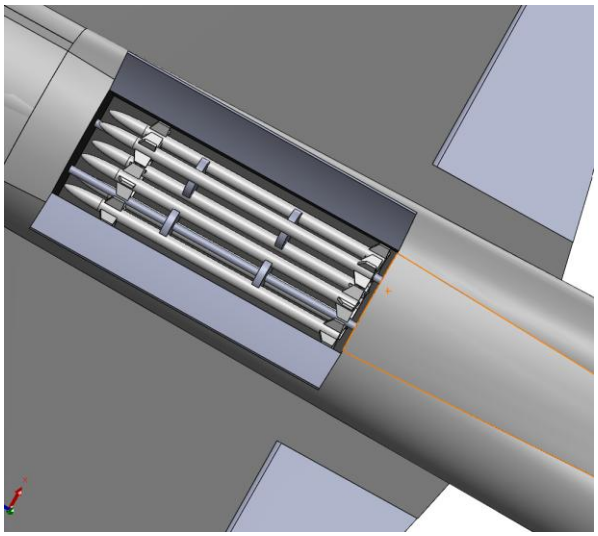


Figure 3-10: Payload Placement

3.2.2.9 Engine Placement

The dual engine configuration is positioned aft of the aircraft fuselage to maximize internal layout efficiency. Positioning the engines at the rear frees up fuselage space for internal payload systems as well as structural components that are essential for aircraft safety and high-G maneuvers. The inlet to these engines lofts up and over the internal payload system to reach the engine intake while still having a smooth flow path. The rear layout also allows for direct thrust generation by having efficient exhaust flow behind the aircraft. From a stability standpoint, placing the engines at the rear is favorable for aircraft center of gravity as the CG shifts are limited as fuel is consumed. The CG will still shift forward when fuel is consumed, which is ideal. Additionally, the dual engine configuration drastically increases survivability in case of single engine failure, allowing the pilots to return safely to the carrier.

3.2.2.10 Landing Gear Type and Placement

The fighter uses oleo-pneumatic struts to manage the forces that will be imposed on the aircraft during takeoff and landing. The aircraft will have one nose landing gear located near the front of the aircraft with two tires and two main landing gears located under the wings of the aircraft with one tire on each strut to help distribute the forces. The landing gear is arranged with the nose gear ahead, and the main gear behind both the forward and aft CG locations. This provides stability for takeoff, landing, and taxiing.

The nose gear will have a launch bar and holdback integrated within the assembly to connect the aircraft to the catapult shuttle on the carrier and hold the aircraft until it is

ready to launch. For a jet aircraft to land on a carrier, it must be able to come to a complete stop within a very short distance so that it will not fall off the carrier. To do so, the aircraft will be equipped with a tail hook that is capable of attaching to the arresting wire on the deck that helps bring the fighter to a complete stop. This tail hook is made from high-tensile strength materials and incorporated within the frame of the aircraft. It will be mounted in-line and on the center of the aircraft to ensure it can reliably snag the arresting wire. The landing gear and tailhook will also be retractable to aid in aerodynamics, reduce drag, and improve stealth. This configuration ensures structural stability and optimized weight distribution while enhancing safety, reliability, and operational performance.

3.3 Configuration and Drawings

3.3.1 General Arrangement 3-View Drawing

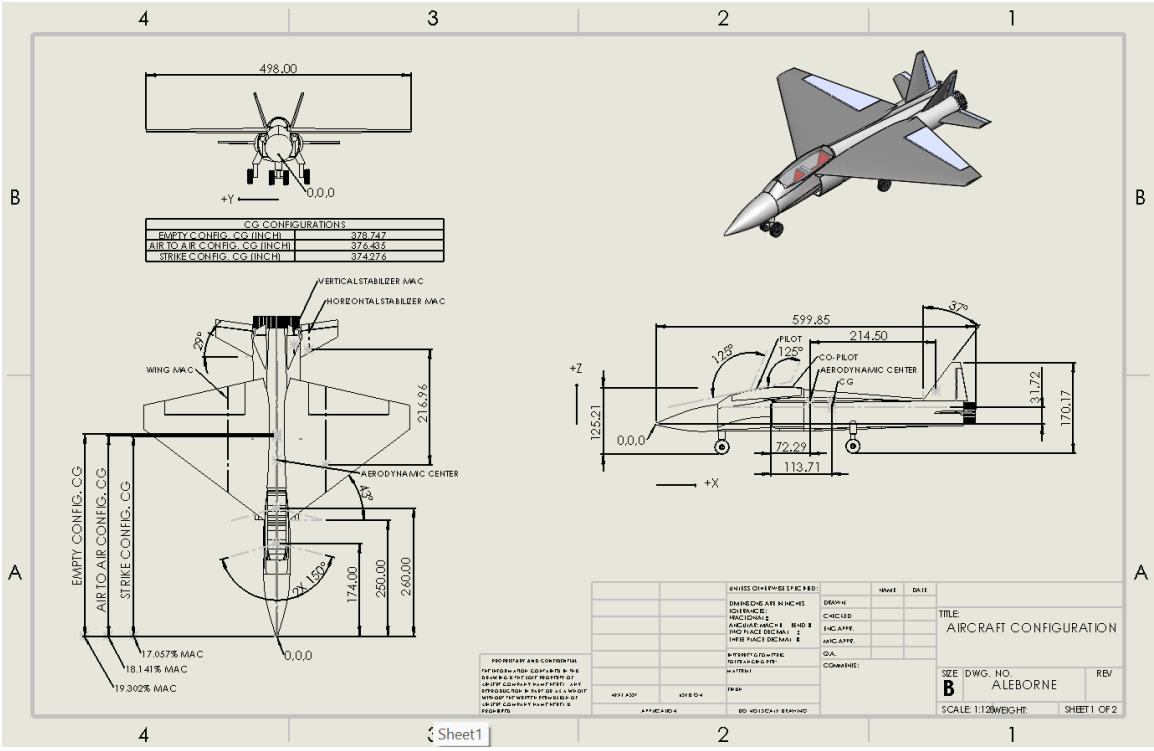
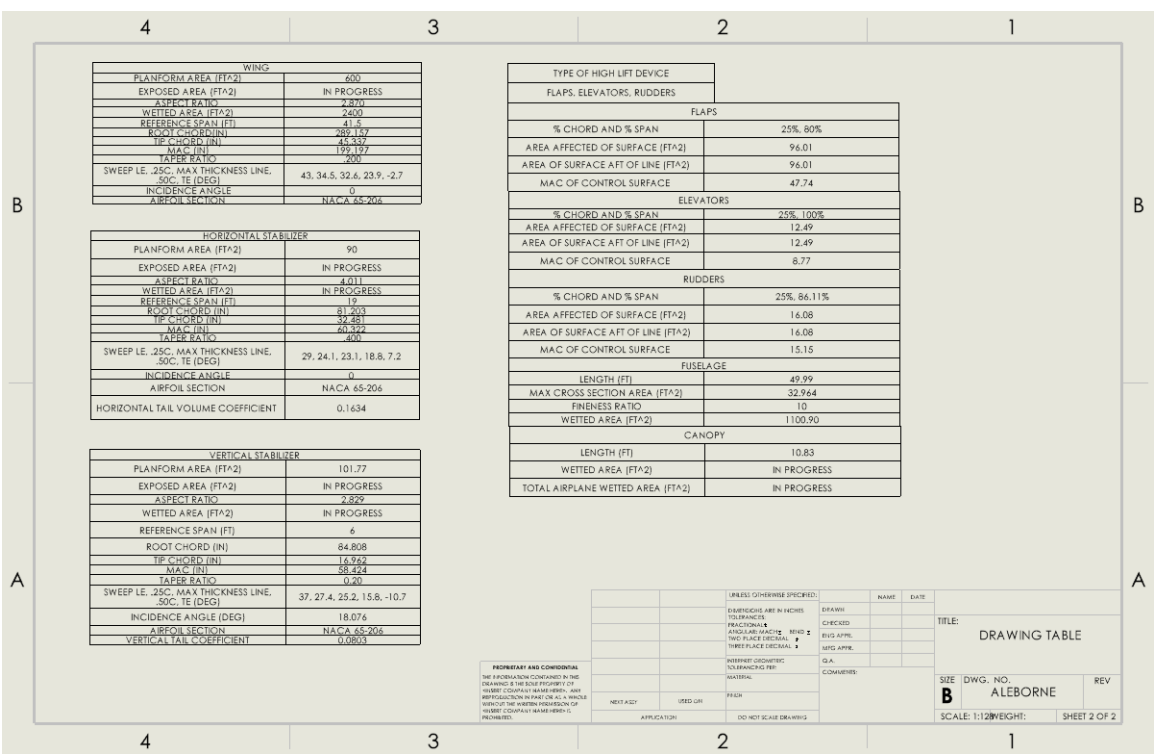


Figure 3-11: General Arrangement 3-View Drawing

3.3.2 Drawing Table



WING	
PLANFORM AREA (FT^2)	600
EXPOSED AREA (FT^2)	IN PROGRESS
ASPECT RATIO	2.870
WETTED AREA (FT^2)	2400
REFERENCE SPAN (FT)	41.5
ROOT CHORD(IN)	289.157
TIP CHORD (IN)	45.337
MAC (IN)	199.197
TAPER RATIO	.200
SWEEP LE, .25C, MAX THICKNESS LINE, .50C, TE (DEG)	43, 34.5, 32.6, 23.9, -2.7
INCIDENCE ANGLE	0
AIRFOIL SECTION	NACA 65-206

HORIZONTAL STABILIZER	
PLANFORM AREA (FT^2)	90
EXPOSED AREA (FT^2)	IN PROGRESS
ASPECT RATIO	4.011
WETTED AREA (FT^2)	IN PROGRESS
REFERENCE SPAN (FT)	19
ROOT CHORD (IN)	81.203
TIP CHORD (IN)	32.481
MAC (IN)	60.322
TAPER RATIO	.400
SWEEP LE, .25C, MAX THICKNESS LINE, .50C, TE (DEG)	29, 24.1, 23.1, 18.8, 7.2
INCIDENCE ANGLE	0
AIRFOIL SECTION	NACA 65-206
HORIZONTAL TAIL VOLUME COEFFICIENT	0.1634

VERTICAL STABILIZER	
PLANFORM AREA (FT^2)	101.77
EXPOSED AREA (FT^2)	IN PROGRESS
ASPECT RATIO	2.829
WETTED AREA (FT^2)	IN PROGRESS
REFERENCE SPAN (FT)	6
ROOT CHORD (IN)	84.808
TIP CHORD (IN)	16.962
MAC (IN)	58.424
TAPER RATIO	0.20
SWEEP LE, .25C, MAX THICKNESS LINE, .50C, TE (DEG)	37, 27.4, 25.2, 15.8, -10.7
INCIDENCE ANGLE (DEG)	18.076
AIRFOIL SECTION	NACA 65-206
VERTICAL TAIL COEFFICIENT	0.0803

TYPE OF HIGH LIFT DEVICE
FLAPS, ELEVATORS, RUDDERS

FLAPS	
% CHORD AND % SPAN	25%, 80%
AREA AFFECTED OF SURFACE (FT^2)	96.01
AREA OF SURFACE AFT OF LINE (FT^2)	96.01
MAC OF CONTROL SURFACE	47.74
ELEVATORS	
% CHORD AND % SPAN	25%, 100%
AREA AFFECTED OF SURFACE (FT^2)	12.49
AREA OF SURFACE AFT OF LINE (FT^2)	12.49
MAC OF CONTROL SURFACE	8.77
RUDDERS	
% CHORD AND % SPAN	25%, 86.11%
AREA AFFECTED OF SURFACE (FT^2)	16.08
AREA OF SURFACE AFT OF LINE (FT^2)	16.08
MAC OF CONTROL SURFACE	15.15
FUSELAGE	
LENGTH (FT)	49.99
MAX CROSS SECTION AREA (FT^2)	32.964
FINENESS RATIO	10
WETTED AREA (FT^2)	1100.90
CANOPY	
LENGTH (FT)	10.83
WETTED AREA (FT^2)	IN PROGRESS
TOTAL AIRPLANE WETTED AREA (FT^2)	IN PROGRESS

Figure 3-12: Drawing Table

3.3.3 Inboard Profile

TBD

4 Mass Properties

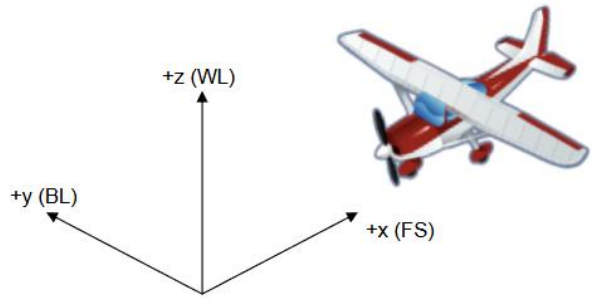


Figure 4-1: Aircraft Coordinate System Conversion

4.1 Detailed Weight Estimate

Major Structural Components

For the Roskam specific estimates for the baseline weights, please refer to figure 3-1 and 3-2.

The preliminary weight estimates are a useful tool to understand if the aircraft proposed is within reason prior to the expenditure of a vast number of resources on the development of the aircraft. Post analysis has determined that this aircraft is not only possible but can be refined to a reasonable weight and have the potential to perform very well. The current iteration of these weight estimates uses Roskam's Part V for major structural components as well as the weight of the fuel. It is the opinion of this team that other methods such as Nicolai's textbooks could potentially produce more satisfactory results; however, this will be analyzed further as the development of the F-44 continues.

To begin the weight estimation process, our team began with the estimated wing weight. This was calculated using Roskam's Part V in the equation 4-1 shown below. A condensed table full of all variables used will be attached to the end of this section for ease of reference.

$$W_{W} = 19.29\left(\left(\frac{(K_w n_{ult} W_{TO})}{\left(\frac{t}{c}\right)_m}\right)\left(\left(\tan(\lambda_{LE}) - \frac{2(1-\lambda)}{A(1+\lambda)}\right)^2 + 1\right)\cdot 10^{-6}\right)^{0.464}$$

$$* \left((1+\lambda)A\right)^{0.7}(S)^{0.58}\left((1+\lambda)A\right)^{0.7}(S)^{0.58}$$

Equation 4-1

Moving onto the empennage, the horizontal stabilizer weight was calculated first similarly using Roskam's Part V, shown in equation 4-2.

$$W_h = 0.0034\left((W_{TO} n_{ult})^{0.813}(S_h)^{0.584}\left(\frac{b_h}{t_{r_h}}\right)^{0.033}\left(\frac{\bar{c}}{l_h}\right)^{0.28}\right)^{0.915}$$

Equation 4-2

And finally, to finish out the major aerodynamic surfaces, the vertical stabilizer was calculated using equation 3-4 shown below.

$$W_v = 0.19\left(\left(1+\frac{z_h}{b_v}\right)^{0.5}(W_{TO} n_{ult})^{0.363}(S_v)^{1.089}(M_H)^{0.601}(l_v)^{-0.726}\left(1+\frac{S_r}{S_v}\right)^{0.217}\right)^{1.014}$$

$$+ 0.19\left((A_v)^{0.337}(1+\lambda_v)^{0.363}\left(\cos\lambda_{\frac{1}{4}w}\right)^{-0.484}\right)^{1.014}$$

Equation 4-3

After the major lift producing and aerodynamic surfaces were calculated, the fuselage was the next large item to produce. The theoretical weight of the fuselage is shown below.

$$W_f = 10.43(K_{inl})^{1.42}\left(\frac{\bar{q}_D}{100}\right)^{0.283}\left(\frac{W_{TO}}{1000}\right)^{0.95}\left(\frac{l_f}{h_f}\right)^{0.71}$$

Equation 4-4

To use Roskam's Part V for the weight assumptions involving the fuselage, a few assumptions are made. Equation (4) is specifically used for commercial transport planes; however, it is outlined that to use this relation, to remove a factor of 2 from the beginning of the equation. While we believe that this theoretical weight will suffice, it is important to note that this may be a broad/conservative estimate, as Roskam Chapter V did not specifically outline their own equation for a fighter aircraft, rather have the reader remove a factor specifically for fighter jet aircraft. Other avenues such as

Nicolai will be investigated as the research and development continues, specifically if weight problems become an issue.

In addition, when calculating the dynamic dive pressure, a landing weight is needed. So, for our purposes, we assume a 25% fuel reserve and 50% payload expenditures for an average mission. Other assumptions include a sea level landing for density and a maximum coefficient of lift that will be achieved during landing. All other parameters such as surface area of the wings and stall velocity are produced in an excel spreadsheet and are currently known values. Shown in Equation 4-5 is the exact method used to collect the dynamic dive pressure.

$$q_d = \frac{1}{2}\boldsymbol{\rho} V_S^2$$

Equation 4-5

Minor components

The following individual components have a much smaller impact on the overall CG of the aircraft but combined can have a surprisingly large total weight. Due to this, a list of important gear, minor structural components, and personnel are collected and calculated individually. For these calculations, Nicolai's Fundamentals Aircraft Airship Design outline relations that can be used to estimate the weight impact for smaller components. These are shown below.

$$W_{LandGear} = 0.054(W_{Land})^{0.501}(W_{Land}N_{ult})^{0.684}$$

Equation 4-6

Next is the propulsion/fuel system. This is responsible for transporting the fuel from the tank to the engine for ignition.

$$W_{PSys} = 2.575\left(W_{Eng}\right)^{0.922}N_E$$

Equation 4-7

For the pilot to maintain control of the control surfaces, engines, payload, and communications, electrical systems must be installed, which have a significant weight value. This is calculated using equation 3-9

$$W_{ElecSys} = 426\left(\frac{(W_{PSys}+W_{Tron})}{1000}\right)^{0.51}$$

Equation 4-8

Equation 3-9 references a variable named W_{Tron},which is the total avionics weight installed on the aircraft. Note that this is different than the control surface systems' weight. The value for W_{Tron} is given using equation 3-10 below.

$$W_{Tron} = 2.117(W_{AU})^{0.933}$$

Equation 4-9

Similarly, W_{Tron} requires a value for W_{AU}, or the uninstalled bare avionics weight. This is given using table 8.7 in Nicolai's design book referenced in the introduction to minor components of weight estimates. It has been decided that a conservative value of 500 lbs is acceptable. This encompasses items such as communication systems, computers, compasses, radar altimeter, attack radar, HUD, etc.

Finally, for the pilot to be able to pilot the plane appropriately, he must have control over all control surfaces installed on the aircraft. A detailed estimate of all involved actuators and control systems is outlined in equation 3-11.

$$W_{SurfCon} = 1.08(W_{TO})^{0.7}$$

Equation 4-10

There are a few items that are essential for crew comfort/safety. Two of which is internal air-conditioning which will serve an additional purpose to de-ice the plane, and the ejection seat for emergency cases. These are shown in Equations 4-12 and 4-13.

$$W_{AC} = 0.265(W_{TO})^{0.52}(N_{CR} + N_{Pass})^{0.68}(W_{Tron})^{0.17}(M_E)^{0.08}$$

Equation 4-11

$$W_{EjecSeat} = 34.5(N_{CR})(q)^{0.25}$$

Equation 4-12

Fuel Weight Estimation

The major components of this aircraft such as fuselage, wings, and empennage are all estimated using Roskam's methods. To remain consistent with the other large weight components, instead of using Nicolai's estimation, Roskam's Part II is employed below. The first requirement for estimating the necessary fuel to complete a mission is the

$$M_{ff} = \prod_{i=1}^{n}\left(\frac{W_i}{W_i - 1}\right)$$

Equation 4-13

Equation 3-13 outlines that the product of all weight fractions for each mission segment is the final value required for the calculation done in equation 3-15. These segments include warmup/takeoff, climb, cruise, combat, loiter, descent, landing, and reserve. Once each fuel fraction is obtained, the product is used below in equation 3-15.

$$W_F = W_{TO}\left(\left(1 - M_{ff}\right) + f_{res}\right)$$

Equation 4-14

Other Notable Weights

These are weights that are either given in the RFP/SRD or are able to be sourced from manufacturers.

For the current design of the F-44 we have elected to pursue a dual engine layout. The selected engine, the F404-GE-402, has a given weight provided by the manufacturer. Because of the wing-fuel storage system, the current weight of the fuel tanks is not calculated as an external store. They are located and factored into the current wing weight. The exact dimension of the fuel stores is currently under review due to frequent wing parameter design changes. As these numbers become finalized, a complete fuel store weight will be provided.

Payload weight for each individual type of store is provided by the manufacturer and outlined.

Crew weight is assumed per the RCR/SRD, no estimations are required.

Trapped fuel/oil is assumed to be a 3-6% fuel fraction margin within Chapter 5 of Nicolai.

4.2 Detailed Weight Breakdown Tables

Shown below is a detailed breakdown of the individual components aboard the F-44. This table consists of derived/calculated values that have been color coded by grouping for ease of use for each mission.

Table 4-1. Detailed weight breakdown of individual components for the strike mission.

Component Name	Weight (lbs)
Wings	3900
Horizontal Stabilizer	480

Component Name	Weight (lbs)
Vertical Stabilizer	950
Fuselage	3500
Nose Landing Gear	100
Rear Landing Gear	200
Propulsion/Fuel System	220
Electronic System	1150
Avionics System	698
Control Surfaces Controllers/System	1890
De-Icing/Air Conditioning	222
Ejection Seat 1	88
Ejection Seat 2	88
Fuel Wing (Left)	9000
Fuel Wing (Right)	9000
Engine (Left)	2282
Engine (Right)	2282
Fuel Tanks	IP
Propulsion Systems Minus Engine Weight	1400
Payload 1, MK-83 JDAM	4000
Payload 2, AIM-9X	375
Crew 1	220
Crew 2	220
Trapped Fuel/Oil	100
Furnishing/Equipment	175

Table 4-2. Detailed weight breakdown of individual components for the A2A mission.

Component Name	Weight (lbs)
Wings	3900
Horizontal Stabilizer	480
Vertical Stabilizer	950
Fuselage	3500
Nose Landing Gear	100
Rear Landing Gear	200

Component Name	Weight (lbs)
Propulsion/Fuel System	220
Electronic System	1150
Avionics System	698
Control Surfaces Controllers/System	1890
De-Icing/Air Conditioning	222
Ejection Seat 1	88
Ejection Seat 2	88
Fuel Wing (Left)	9700
Fuel Wing (Right)	9700
Engine (Left)	2282
Engine (Right)	2282
Fuel Tanks	IP
Propulsion Systems – Engine Weight	1400
Payload 1, AIM-120C	2100
Payload 2, AIM-9X	375
Crew 1	220
Crew 2	220
Trapped Fuel/Oil	100
Furnishing/Equipment	175

Center of Gravity Calculations

Table 4-3. Detailed center of gravity breakdown showcasing the overall CG, total weight, and %MAC of the aircraft for the strike mission.

FULL CONFIG STRIKE				
Component	**Weight (lbs)**	**X (in)**	**Y (in)**	**Z (in)**
Fuel (Wings)	18276	380	0	46
Wings	3900	380	0	46
H-Stab	480	550	0	0
V-Stab	950	550	0	72
Fuselage	3500	230	0	0
Nose Gear	100	220	0	-15
Rear Gear	200	400	0	-15

FULL CONFIG STRIKE				
Component	**Weight (lbs)**	**X (in)**	**Y (in)**	**Z (in)**
Pilot	220	210	0	59
Nose Cone	220	80	0	0
Canopy	300	210	0	59
Seating	100	210	0	16
Thrust Nozzle (Left)	400	591	23	20
Thrust Nozzle (Right)	400	591	-23	20
Engine 1 (Left)	2282	488	23	20
Engine 2 (Right)	2282	488	-23	20
Payload System 1 (Left)	2187	330	13	13
Payload System 2 (Right)	2187	330	-13	13
Propulsion Systems (TOTAL)	6429	488	-23	20
Avionics System	698	212.5	0	35
Control Surface System Wings	1892	380	0	46
Control Surface System Empennage	210	550	0	40
Flight Equipment (Pilot 1)	25	180	0	35
Flight Equipment (Pilot 2)	25	245	0	35
Ejection Seat 1	87	180	0	35
Ejection Seat 2	87	245	0	35
De-Icing/AC System	222	300	0	0
Electrical System	1159	360	0	0
Furnishing/Equipment	175	210	0	45
Total Weight (lbs)	**CG Location, FS (in)**	**%MAC**		
42567	374	17.05		

Table 4-4. Detailed center of gravity breakdown showcasing the overall CG, total weight, and %MAC of the aircraft for the A2A mission.

FULL CONFIG. A2A				
		Direction (in)		
Component	**Weight (lbs)**	**X**	**Y**	**Z**
Fuel (Wings)	19307	380	0	46
Wings	3900	380	0	46
H-Stab	480	550	0	0
V-Stab	950	550	0	72
Fuselage	3500	230	0	0

FULL CONFIG. A2A				
Nose Gear	100	220	0	-15
Rear Gear	200	400	0	-15
Pilot	220	210	0	59
Nose Cone	220	80	0	0
Canopy	300	210	0	59
Seating	100	210	0	16
Thrust Nozzle (Left)	400	591	23	20
Thrust Nozzle (Right)	400	591	-23	20
Engine 1 (Left)	2282	488	23	20
Engine 2 (Right)	2282	488	-23	20
Payload System 1 (Left)	1237	330	13	13
Payload System 2 (Right)	1237	330	-13	13
Propulsion Systems (TOTAL)	6429	488	-23	20
Avionics System	698	212.5	0	35
Control Surface System Wings	1892	380	0	46
Control Surface System Empennage	210	550	0	40
Flight Equipment (Pilot 1)	25	180	0	35
Flight Equipment (Pilot 2)	25	245	0	35
Ejection Seat 1	87	180	0	35
Ejection Seat 2	87	245	0	35
De-Icing/AC System	222	300	0	0
Electrical System	1159	360	0	0
Furnishing/Equipment	175	210	0	45
Total Weight (lbs)	**CG Location, FS (in)**		**%MAC**	
41699	376		18.1	

Table 4-5. Detailed center of gravity breakdown showcasing the overall CG, total weight, and %MAC of the aircraft for an empty configuration.

EMPTY CONFIG				
		Direction (in)		
Component	**Weight (lbs)**	**X**	**Y**	**Z**
Fuel (Wings)	0	380	0	46
Wings	3900	380	0	46
H-Stab	480	550	0	0
V-Stab	950	550	0	72

EMPTY CONFIG				
Fuselage	3500	230	0	0
Nose Gear	100	220	0	-15
Rear Gear	200	400	0	-15
Pilot	220	210	0	59
Nose Cone	220	80	0	0
Canopy	300	210	0	59
Seating	100	210	0	16
Thrust Nozzle (Left)	400	591	23	20
Thrust Nozzle (Right)	400	591	-23	20
Engine 1 (Left)	2282	488	23	20
Engine 2 (Right)	2282	488	-23	20
Payload System 1 (Left)	0	330	13	13
Payload System 2 (Right)	0	330	-13	13
Propulsion Systems (TOTAL)	6429	488	-23	20
Avionics System	698	212.5	0	35
Control Surface System Wings	1892	380	0	46
Control Surface System Empennage	210	550	0	40
Flight Equipment (Pilot 1)	25	180	0	35
Flight Equipment (Pilot 2)	25	245	0	35
Ejection Seat 1	87	180	0	35
Ejection Seat 2	87	245	0	35
De-Icing/AC System	222	300	0	0
Electrical System	1159	360	0	0
Furnishing/Equipment	175	210	0	45

Total Weight (lbs)	CG Location, FS (in)	%MAC
41699	376	18.1

Table 4-6. Detailed center of gravity breakdown showcasing the overall CG, total weight, and %MAC of the aircraft for the clean cruise configuration.

Clean Cruise Config.				
		Direction (in)		
Component	**Weight (lbs)**	**X**	**Y**	**Z**
Fuel (Wings)	10965	380	0	46
Wings	3900	380	0	46
H-Stab	480	550	0	0

Clean Cruise Config.				
V-Stab	950	550	0	72
Fuselage	3500	230	0	0
Nose Gear	100	220	0	-15
Rear Gear	200	400	0	-15
Pilot	220	210	0	59
Nose Cone	220	80	0	0
Canopy	300	210	0	59
Seating	100	210	0	16
Thrust Nozzle (Left)	400	591	23	20
Thrust Nozzle (Right)	400	591	-23	20
Engine 1 (Left)	2282	488	23	20
Engine 2 (Right)	2282	488	-23	20
Payload System 1 (Left)	0	330	13	13
Payload System 2 (Right)	0	330	-13	13
Propulsion Systems (TOTAL)	6429	488	-23	20
Avionics System	698	212.5	0	35
Control Surface System Wings	1892	380	0	46
Control Surface System Empennage	210	550	0	40
Flight Equipment (Pilot 1)	25	180	0	35
Flight Equipment (Pilot 2)	25	245	0	35
Ejection Seat 1	87	180	0	35
Ejection Seat 2	87	245	0	35
De-Icing/AC System	222	300	0	0
Electrical System	1159	360	0	0
Furnishing/Equipment	175	210	0	45
Total Weight (lbs)	**CG Location, FS (in)**		**%MAC**	
30883	379.2		19.52	

Table 4-7. Detailed center of gravity breakdown showcasing the overall CG, total weight, and %MAC of the aircraft for the ferry cruise configuration.

Ferry Cruise Config.				
		Direction (in)		
Component	**Weight (lbs)**	**X**	**Y**	**Z**
Fuel (Wings)	18276	380	0	46
Wings	3900	380	0	46

Ferry Cruise Config.				
		Direction (in)		
Component	**Weight (lbs)**	**X**	**Y**	**Z**
H-Stab	480	550	0	0
V-Stab	950	550	0	72
Fuselage	3500	230	0	0
Nose Gear	100	220	0	-15
Rear Gear	200	400	0	-15
Pilot	220	210	0	59
Nose Cone	220	80	0	0
Canopy	300	210	0	59
Seating	100	210	0	16
Thrust Nozzle (Left)	400	591	23	20
Thrust Nozzle (Right)	400	591	-23	20
Engine 1 (Left)	2282	488	23	20
Engine 2 (Right)	2282	488	-23	20
Payload System 1 (Left)	0	330	13	13
Payload System 2 (Right)	0	330	-13	13
Propulsion Systems (TOTAL)	6429	488	-23	20
Avionics System	698	212.5	0	35
Control Surface System Wings	1892	380	0	46
Control Surface System Empennage	210	550	0	40
Flight Equipment (Pilot 1)	25	180	0	35
Flight Equipment (Pilot 2)	25	245	0	35
Ejection Seat 1	87	180	0	35
Ejection Seat 2	87	245	0	35
De-Icing/AC System	222	300	0	0
Electrical System	1159	360	0	0
Furnishing/Equipment	175	210	0	45
Total Weight (lbs)	**CG Location, FS (in)**		**%MAC**	
38193	377		18.6	

Table 4-8 Detailed center of gravity breakdown showcasing the overall CG, total weight, and %MAC of the aircraft for the clean weapons configuration.

Weapons Cruise Config.	
	Direction (in)

Component	Weight (lbs)	X	Y	Z
Fuel (Wings)	10965	380	0	46
Wings	3900	380	0	46
H-Stab	480	550	0	0
V-Stab	950	550	0	72
Fuselage	3500	230	0	0
Nose Gear	100	220	0	-15
Rear Gear	200	400	0	-15
Pilot	220	210	0	59
Nose Cone	220	80	0	0
Canopy	300	210	0	59
Seating	100	210	0	16
Thrust Nozzle (Left)	400	591	23	20
Thrust Nozzle (Right)	400	591	-23	20
Engine 1 (Left)	2282	488	23	20
Engine 2 (Right)	2282	488	-23	20
Payload System 1 (Left)	2282	330	13	13
Payload System 2 (Right)	2282	330	-13	13
Propulsion Systems (TOTAL)	6429	488	-23	20
Avionics System	698	212.5	0	35
Control Surface System Wings	1892	380	0	46
Control Surface System Empennage	210	550	0	40
Flight Equipment (Pilot 1)	25	180	0	35
Flight Equipment (Pilot 2)	25	245	0	35
Ejection Seat 1	87	180	0	35
Ejection Seat 2	87	245	0	35
De-Icing/AC System	222	300	0	0
Electrical System	1159	360	0	0
Furnishing/Equipment	175	210	0	45

Total Weight (lbs)	CG Location, FS (in)	%MAC
35257	373	16.46

Table 4-9. Detailed center of gravity breakdown showcasing the overall CG, total weight, and %MAC of the aircraft for the weapons take-off configuration.

Weapons Take-Off Config.	
	Direction (in)

Component	Weight (lbs)	X	Y	Z
Fuel (Wings)	18276	380	0	46
Wings	3900	380	0	46
H-Stab	480	550	0	0
V-Stab	950	550	0	72
Fuselage	3500	230	0	0
Nose Gear	100	220	0	-54
Rear Gear	200	400	0	-54
Pilot	220	210	0	59
Nose Cone	220	80	0	0
Canopy	300	210	0	59
Seating	100	210	0	16
Thrust Nozzle (Left)	400	591	23	20
Thrust Nozzle (Right)	400	591	-23	20
Engine 1 (Left)	2282	488	23	20
Engine 2 (Right)	2282	488	-23	20
Payload System 1 (Left)	2282	330	13	13
Payload System 2 (Right)	2282	330	-13	13
Propulsion Systems (TOTAL)	6429	488	-23	20
Avionics System	698	212.5	0	35
Control Surface System Wings	1892	380	0	46
Control Surface System Empennage	210	550	0	40
Flight Equipment (Pilot 1)	25	180	0	35
Flight Equipment (Pilot 2)	25	245	0	35
Ejection Seat 1	87	180	0	35
Ejection Seat 2	87	245	0	35
De-Icing/AC System	222	300	0	0
Electrical System	1159	360	0	0
Furnishing/Equipment	175	210	0	45

Total Weight (lbs)	CG Location, FS (in)	%MAC
42567	374	17.1

Table 4-10. Detailed center of gravity breakdown showcasing the overall CG, total weight, and %MAC of the aircraft for the weapons landing configuration.

Weapons Landing Config.	
	Direction (in)

Component	Weight (lbs)	X	Y	Z
Fuel (Wings)	4569	380	0	46
Wings	3900	380	0	46
H-Stab	480	550	0	0
V-Stab	950	550	0	72
Fuselage	3500	230	0	0
Nose Gear	100	220	0	-54
Rear Gear	200	400	0	-54
Pilot	220	210	0	59
Nose Cone	220	80	0	0
Canopy	300	210	0	59
Seating	100	210	0	16
Thrust Nozzle (Left)	400	591	23	20
Thrust Nozzle (Right)	400	591	-23	20
Engine 1 (Left)	2282	488	23	20
Engine 2 (Right)	2282	488	-23	20
Payload System 1 (Left)	2282	330	13	13
Payload System 2 (Right)	2282	330	-13	13
Propulsion Systems (TOTAL)	6429	488	-23	20
Avionics System	698	212.5	0	35
Control Surface System Wings	1892	380	0	46
Control Surface System Empennage	210	550	0	40
Flight Equipment (Pilot 1)	25	180	0	35
Flight Equipment (Pilot 2)	25	245	0	35
Ejection Seat 1	87	180	0	35
Ejection Seat 2	87	245	0	35
De-Icing/AC System	222	300	0	0
Electrical System	1159	360	0	0
Furnishing/Equipment	175	210	0	45

Total Weight (lbs)	CG Location, FS (in)	%MAC
28860	372	15.69

4.3 Longitudinal CG Travel

At the core of CG analysis is the objective to ensure that as weights change in flight due to payload expenditures and fuel burn, the aircraft will not become violently unstable, putting the pilot and the aircrafts survivability in jeopardy. As a further precaution, a table

for each mission type has been produced to analyze the CG drift as the payload, fuel, and landing gear deployment changes. These are shown below in table 4-11.

Table 4-11. Longitudinal CG travel for the strike mission.

Payload	Gear	% Fuel	FS (in)	%MAC	W_Fuel (lbs)
Full Payload	Up	1	370.044028	14.93271	182.76
(4374 lbs)		60	373.0889112	16.46129	10965.6
		100	374.2757994	17.05713	18276
No Payload	Down	1	378.757988	19.30726	182.76
(0 lbs)		60	379.1916371	19.52496	10965.6
		100	379.3463614	19.60263	18276

Table 4-12. Longitudinal CG travel for the A2A mission.

Payload	Gear	% Fuel	FS (in)	%MAC	W_Fuel (lbs)
Full Payload	Up	1	373.4174112	16.62621	193.07
(4374		60	375.6243767	17.73414	11584.2
		100	376.4347677	18.14097	19307
No Payload	Down	1	378.7586247	19.30758	193.07
(0 lbs)		60	379.207511	19.53293	11584.2
		100	379.363542	19.61126	19307

Table 4-13. Summary of variables used throughout the section

Summary of Variables		
Symbol	**Location**	**Description**
W_W	3-2	Wing Weight
K_w	3-2	Wing Weight Correction Factor
n_{ult}	3-2	Ultimate Load Factor
W_{TO}	3-2	Take-Off Weight
$\frac{t}{c}$	3-2	Maximum Thickness to Chord Ratio of the Wing
λ_{LE}	3-2	Leading Edge Sweep Angle
λ	3-2	Taper Ratio
A	3-2	Aspect Ratio
S	3-2	Wing Surface Area

Summary of Variables		
Symbol	**Location**	**Description**
W_h	3-3	Horizontal Stab. Weight
S_h	3-3	Horizontal Stab. Surface Area
b_h	3-3	Horizontal Stab. Span
t_{rh}	3-3	Thickness at Root of Horizontal Stab.
$\overline{c}_h$	3-3	MAC of Horizontal Stab.
l_h	3-3	Horizontal Stab. Moment Arm
W_v	3-4	Vertical Stab. Weight
z_h	3-4	Vertical Distance Between Aircraft CG and V-Stab.l AC
b_v	3-4	Vertical Stab. Span
M	3-4	Mach Number
l_v	3-4	Vertical Stab. Moment Arm
S_r	3-4	Rudder Area
$\lambda_{\frac{1}{4}}$	3-4	¼ Chord Sweep Angle
$\overline{q}_D$	3-5	Dynamic Dive Pressure
N_E	3-8	Number of Engines
N_{PASS}	3-12	Number of Passengers
N_{CR}	3-12	Number of Creq

5 Flight Loads (AE 460B only)

TBD

6 Structures (AE 460B only)

TBD

7 Lift, Drag and Moment Estimation

The aircraft lift was estimated using the methodology outlined by Dr. Jan Roskam. Initially, lift was calculated for an idealized two-dimensional airfoil section. Then, the lift curve slope was expanded to the wing which was used to estimate the lift for the entirety of the aircraft.

The airfoil lift-curve slope at subsonic speeds was estimated using a correction factor, as shown in Equation 7-1:

$$Cl_\alpha = \frac{Cl_\alpha}{(1 - M^2)^{1/2}}$$

Equation 7-1

For this analysis, a Mach number of *M = 0.*1 was assumed. This value was selected as a guiding parameter because it is representative of the low-speed flight regime governing takeoff performance and lift requirements. At this Mach number, compressibility effects are minimal; however, the correction was retained for consistency and completeness.

Although a full supersonic aerodynamic analysis was not performed, an appropriate compressibility correction can also be applied in the supersonic regime. Under supersonic conditions, the lift-curve slope may be approximated as

$$Cl_\alpha = \frac{Cl_\alpha}{(M^2 - 1)^{1/2}}$$

Equation 7-2

Using these relations, a representative lift-curve slope for the airfoil was constructed across the relevant Mach number range.

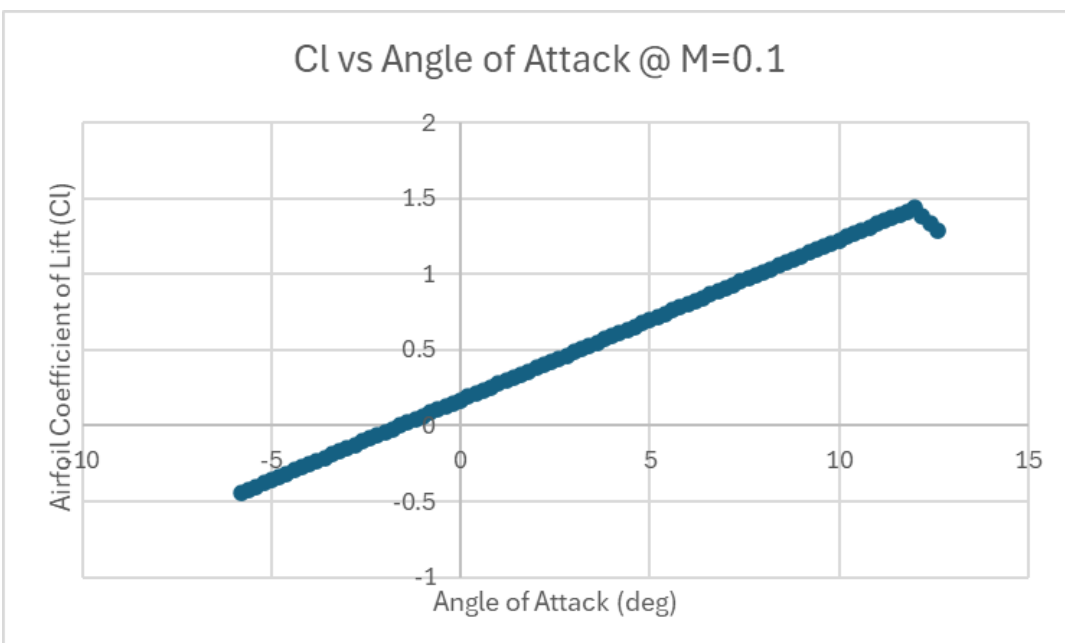


Figure 7-2. Airfoil Coefficient of Lift vs. Angle of Attack at Mach = 0.1

An additional analysis was performed for the airfoil equipped with plain flaps. In this analysis, empirical corrections were applied to account for changes in the lift-curve intercept, lift-curve slope, and maximum lift coefficient. These corrections are based on classical flap theory and semi-empirical correlations commonly used in preliminary aircraft design, as presented by Roskam. The corresponding relations are given in Equations 7-3 to 7-5, respectively.

$$\Delta C_l = (\delta_f)\left\{\frac{C_{l\delta}}{(C_{l\delta})_{theory}}\right\}(C_{l\delta})k'$$

Equation 7-3

$$(\Delta C_\alpha)_\delta = \left(\frac{c'}{c}\right)C_{l\alpha}$$

Equation 7-4

$$\Delta C_{l,max} = k_1 k_2 k_3 (\Delta C_{l,max})_{base}$$

Equation 7-5

With these corrections applied, an updated lift-curve slope was generated using the same assumed Mach number of *M = 0.1*.

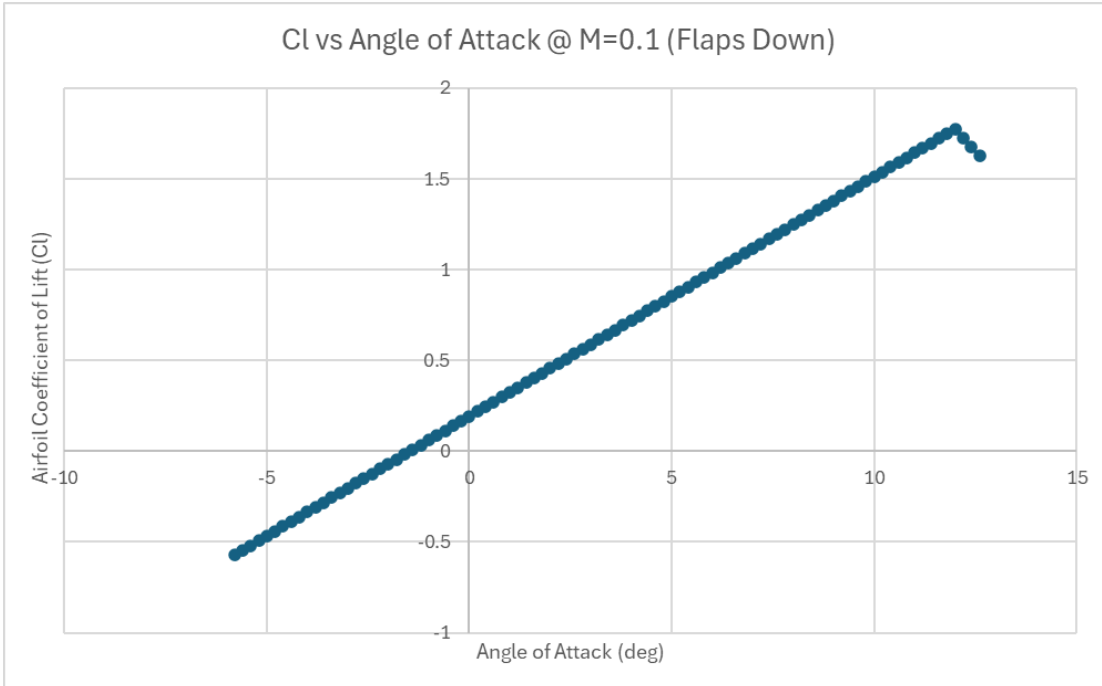


Figure 7-3. Lift Curve Slope for NACA 65-206 at Mach = 0.1 with flaps deployed

To estimate the lift-curve slope of the three-dimensional wing, wing strip theory was employed. In this approach, the total lift coefficient is obtained by integrating the sectional lift contributions along the span. The general formulation is given by

$$C_L = \frac{2}{S}\int_0^{b/2} c(y)c_l(y)dy$$

Equation 7-6

where the sectional lift coefficient is expressed as

$$c_l(y) = a_{0,eff}(y)\alpha_{eff}(y) = a_0(y)\cos\Lambda\big(\alpha_W\cos\Lambda + \varepsilon(y) - \alpha_{L0}(y)\big)$$

Equation 7-7

and the local chord distribution for a linearly tapered wing is

$$c(y) = c_r + (c_t - c_r)\eta = c_r + (c_t - c_r)\frac{2y}{b}$$

Equation 7-8

After integrating across the wingspan, the lift-curve slope of the clean, three-dimensional wing was obtained. This resulting lift-curve slope accounts for the effects of finite wing geometry, including sweep, taper, and spanwise variation in sectional lift characteristics.

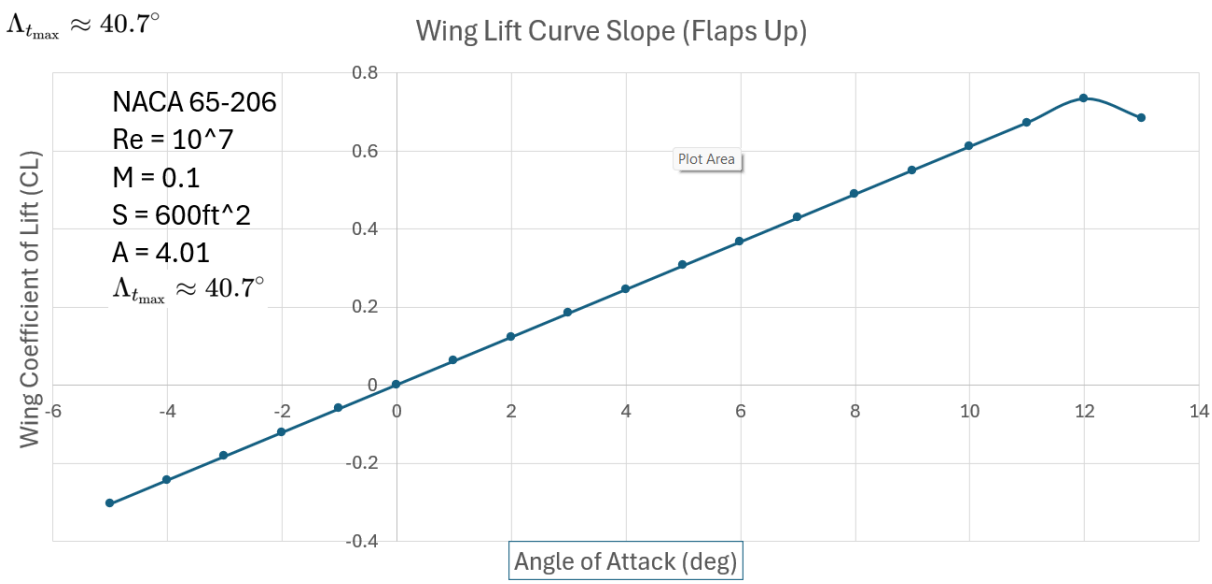


Figure 7-4. Wing Lift Curve Slope for NACA 65-206 at Mach = 0.1 flaps up

The maximum lift coefficient for the aircraft in its clean (flaps retracted) configuration is approximately 0.76, which is sufficient once the aircraft has reached cruise velocity. Based on the calculated three-dimensional wing lift-curve slope, the corresponding angle of attack during cruise is estimated to be approximately –0.58 degrees.

To verify the accuracy of the wing-strip analysis, Roskam's results were cross-checked against those obtained using Pamadi's methodology, which incorporates Datcom-based corrections, as expressed in Equation 7-9:

$$\alpha_{L_{\alpha W}} = \frac{2\pi A}{\left[2 + \left\{\left(\frac{A^2\beta^2}{k^2}\right)\left(1 + \frac{\tan^2 \Lambda_{c/2}}{\beta^2}\right) + 4\right\}^{1/2}\right]}$$

Equation 7-9

Where $\beta = \sqrt{1 - M^2}$ and k is a correction factor that accounts for the wings lift-curve slope.

The results of this comparison are summarized in Table 7-3, demonstrating that the lift-curve slope calculated via wing-strip theory is in reasonable agreement with the Datcom-based approach, providing confidence in the validity of the preliminary aerodynamic estimates for the clean wing configuration.

Table 7-3. Lift coefficient error comparison between Pamadi and Roskam

Angle of Attack (Degrees)	**Pamadi C_L**	**Roskam C_L**	**Error Margin (%)**
-3	-0.1832	-0.1660	10.36%
1	0.0610	0.0398	53.40%
3	0.1832	0.1427	28.37%
6	0.3664	0.2971	23.34%
9	0.5500	0.4514	21.75%
12	0.7328	0.6058	20.97%

As with the two-dimensional plain flap analysis, the wing plain flap analysis accounts for modifications in the wing lift increment, the wing lift-curve slope, and the maximum lift increment due to flap deflection, as reflected in the equations below.

$$\Delta C_{LW} = K_b(\Delta C_l)\left(\frac{\left(C_{L\alpha_W}\right)}{C_{L\alpha}}\right)\left[\frac{(\alpha_\delta)_{C_L}}{(\alpha_\delta)_{C_l}}\right]$$

Equation 7-10

$$(C_{L\alpha W})_\delta = (C_{L\alpha W})\left(\left(1 + \left(\frac{c'}{c} - 1\right)\left(\frac{S_{Wf}}{S}\right)\right)\right)$$

Equation 7-11

$$(\Delta C_{L\max W}) = (\Delta C_{lmax})\left(\frac{S_{Wf}}{S}\right)K_\Delta$$

Equation 7-12

By incorporating the three-dimensional effects, such as finite wingspan, taper, and sweep, the lift-curve slope can be recalculated to reflect the wing's actual aerodynamic performance. This three-dimensional slope captures the reduction in lift efficiency compared to the two-dimensional airfoil and provides a more realistic input for performance calculations, including takeoff, climb, and maneuvering analyses.

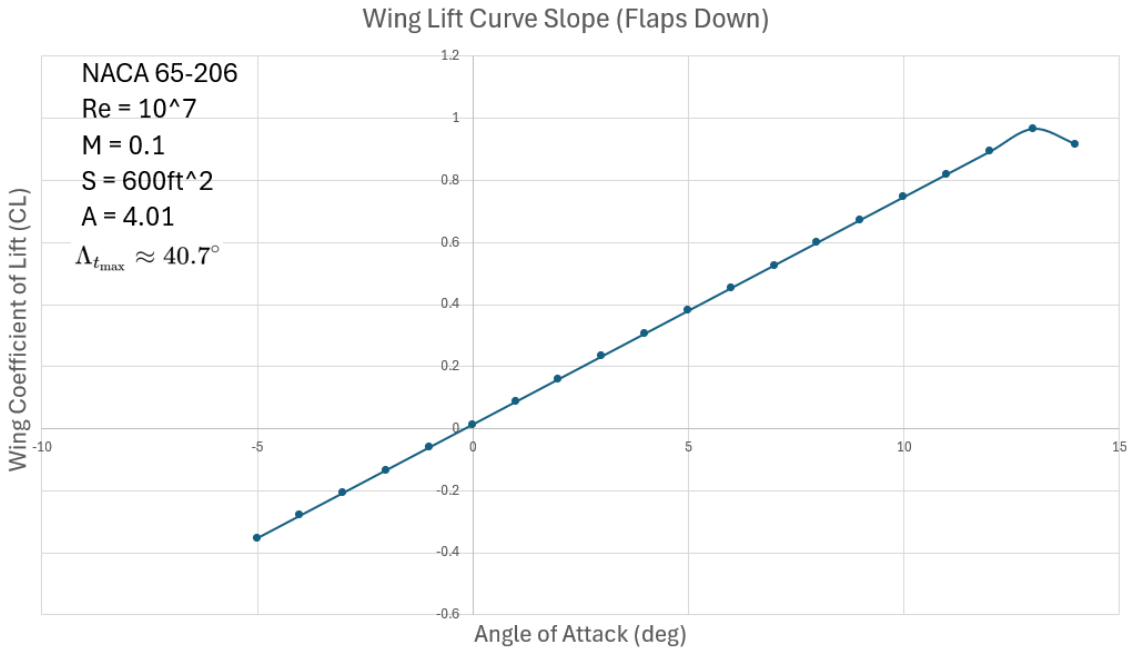


Figure 7-5. Lift Curve Slope using NACA 65-206 airfoil

It should be noted that the maximum lift coefficient for the aircrafts wing with fully plain flaps deployed is ~0.96 which is close to what the actual value will be. This is the value that will be used during takeoff.

The induced-drag factor k is evaluated using the standard relation given in Equation 7-13:

$$k = \frac{1}{\pi e A}$$

Equation 7-13

The Oswald efficiency factor, e, is determined using the empirical correlation proposed by Howe, as shown in Equations 7-14 and 7-15.

$$e = \frac{1}{(1+0.12M^2)\left[1+\frac{0.142+f(\lambda)A(10t/c)^{0.33}}{\left(\cos\Lambda_{c/4}\right)^2}+\frac{0.1(3N_e+1)}{(4+A)^{0.8}}\right]}$$

Equation 7-14

where

$$f(\lambda) = 0.005[1+1.5(\lambda-0.6)^2]$$

Equation 7-15

The Oswald efficiency factor at a Mach number of 1 is calculated as follows:

$$f(0.196) = 0.005[1+1.5(0.196-0.6)^2]$$

$$f(0.196) = 0.00622$$

$$e = \frac{1}{(1+0.12(0.1)^2)\left[1+\frac{0.142+(0.00622)4\left(10(0.06)\right)^{0.33}}{(\cos(0.17524))^2}+\frac{0.1(3(2)+1)}{(4+4)^{0.8}}\right]}$$

$$e = 0.77088$$

$$k = \frac{1}{\pi(0.77088)(4)}$$

$$k = 0.10323$$

The results and corresponding graphs for the Mach number range 0 ≤ M≤ 1.2 are shown in Table 7-4.

Table 7-4. Efficiency Parameters due to Mach number

Mach Number (M)	**Oswald Efficiency Factor (e)**	**Induced Drag (k)**
0.1	0.77088	0.10323
0.2	0.76812	0.1036
0.3	0.76456	0.10422

Mach Number (M)	Oswald Efficiency Factor (e)	Induced Drag (k)
0.4	0.75727	0.10509
0.5	0.74933	0.1062
0.6	0.73985	0.10756
0.7	0.72894	0.10917
0.8	0.71676	0.11102
0.9	0.70343	0.11313
1.0	0.68911	0.11548
1.1	0.67395	0.11808
1.2	0.65803	0.12092

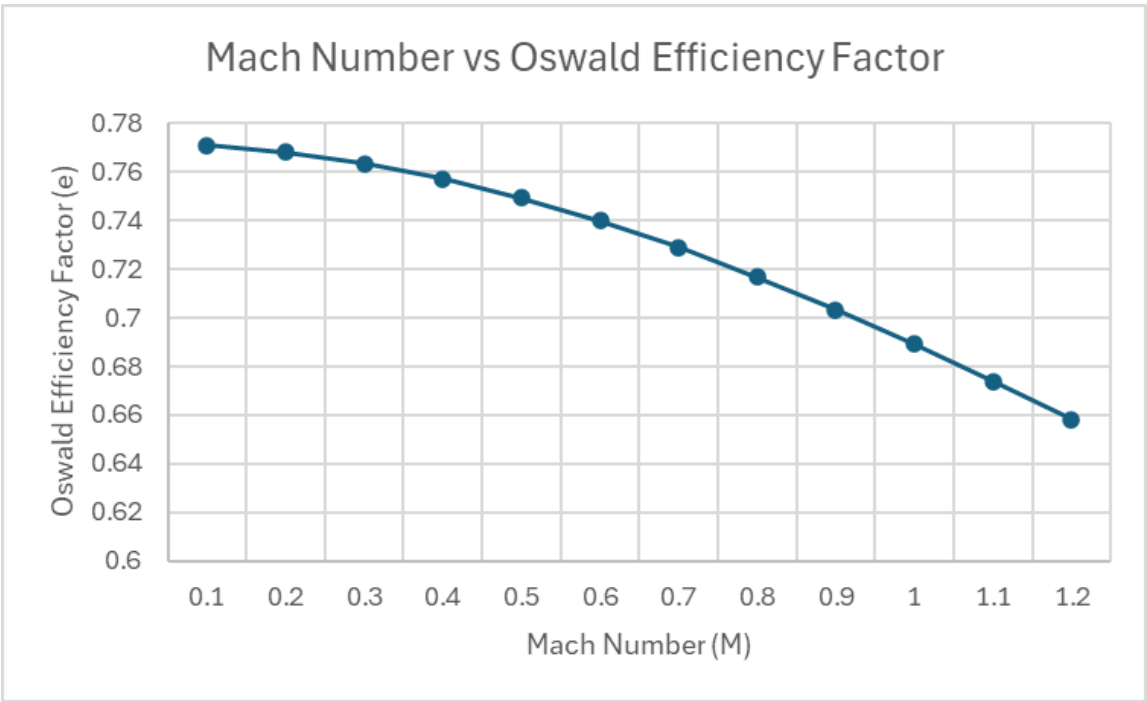


Figure 7-6. Oswald Efficiency Factor vs. Mach Number

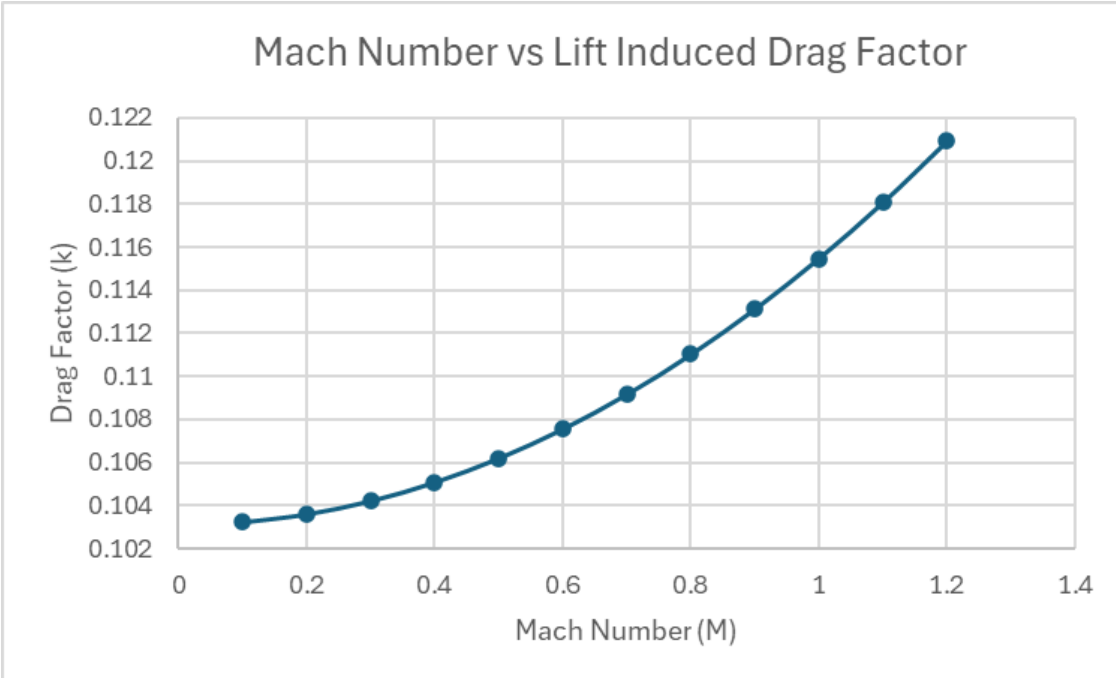


Figure 7-7. Drag Factor vs. Mach Number

The drag polar was estimated using the zero-lift drag coefficients of the fuselage, wing, and empennage. This approach provides a reasonable approximation for preliminary analysis; however, future studies should include corrections for engine nacelles, payload, and other components to refine the drag estimates.

The overall drag polar is expressed as:

$$C_D = C_{D_0} + kC_L^2$$

Equation 7-16

where the zero-lift drag coefficient is the sum of the contributions from the major components:

$$C_{D_0} = C_{D_{0,fus}} + C_{D_{0,emp}} + C_{D_{0,wing}} + C_{D_{wave}}$$

Equation 7-17

This formulation accounts for the primary sources of parasitic drag, including wave drag at higher speeds, while representing lift-induced drag through the quadratic term, kC_L^2.

To estimate the zero-lift drag of the wing under subsonic conditions, Roskam provides the following relation 7-18:

$$C_{D_{0_W}} = (R_{wf})(R_{LS})(Cf_W)(1 + L'(\frac{t}{c}) + 100(\frac{t}{c})^4)\ \frac{S_{wet_W}}{S}$$

Equation 7-18

Where $\frac{S_{wet_W}}{S}$ is the ratio of the wing's wetted area to its planform area and the rest of the coefficients. This formulation captures the primary parasitic drag contribution of the wing at low subsonic speeds.

Using the known aircraft parameters, the zero-lift drag coefficient of the wing at $M = 0.25$ can be calculated as follows:

$$C_{D_{0_W}} = (1.05)(1)(0.0031)(1 + 2(0.06) + 100(0.06)^4)(4)$$

$$C_{D_{0_W}} = 0.0052$$

A similar methodology can be applied to estimate the zero-lift drag coefficient in the supersonic regime, with the addition of a wave-drag term to account for compressibility effects at higher Mach numbers. For a wing with a subsonic leading edge, the wave drag can be expressed as:

$$C_{Dw_{wave}} = C_{D_{LE}} + (\frac{16}{3})cot(\Lambda_{LE_{bw}})((\frac{t}{c})_{eff})^2 \frac{S_{bw}}{S}$$

Equation 7-19

The equation is evaluated at the Mach number where wave-drag effects become significant, which is taken here as $M = 0.8$:

$$C_{Dw_{wave}} = \left(\frac{16}{3}\right)\cot(42°)\,(0.06)^2(0.85)$$

$$C_{DW_{wave}} = 0.0002$$

Using a similar empirical approach, the zero-lift drag of the fuselage can be estimated through correlations that account for fuselage geometry, wetted area, and Reynolds

number effects 7-19. These relations provide a first-order approximation of the fuselage contribution to the total parasitic drag.

$$C_{D0_{fus}} = R_{wf} C_{f_{fus}} \left(1 + \frac{60}{\left(\frac{l_f}{d_f}\right)^3} + 0.0025 \left(\frac{l_f}{d_f}\right) \right) \frac{S_{wet_{fus}}}{S} + C_{Db_{fus}}$$

Equation 7-20

With our knowns, we can estimate our fuselage drag at $M = 0.25$ as follows:

$$C_{D0_{fus}} = (1.05)(0.0029)\left(1 + \frac{60}{(10)^3} + 0.0025(10)\right)(0.23) + (0.02)$$

$$C_{D0_{fus}} = 0.2076$$

Additionally, the supersonic drag of the fuselage can be estimated using the correlation shown in the figure below.

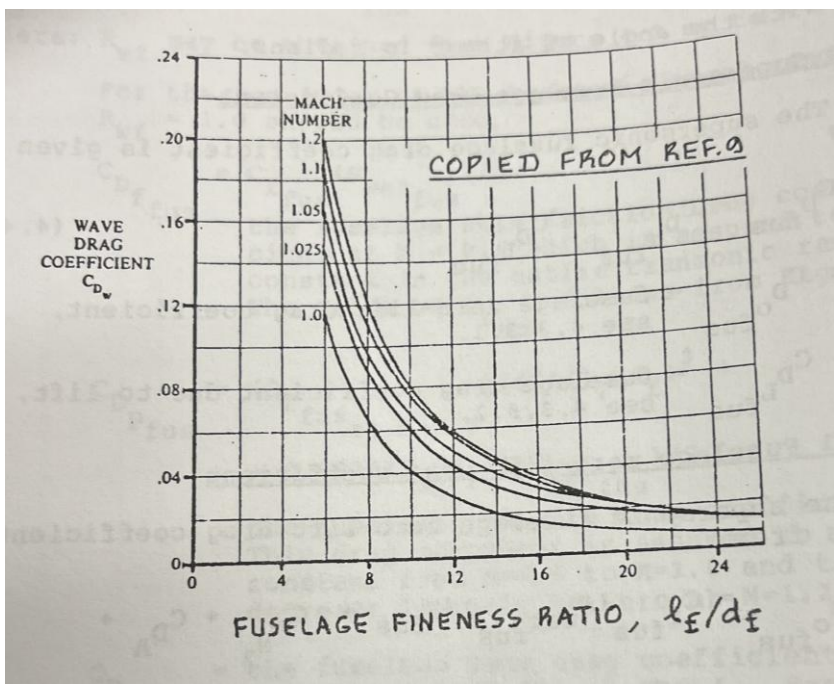


Figure 7-8. Wave Drag Coefficient for Parabolic Fuselages

From Inspection of Figure 7-8, we can assume the wave drag of the fuselage at $M = 0.8$ is roughly $C_{D_{0wave}} = 0.06$.

Finally, the zero-lift drag of the empennage can be calculated using the same method applied to the wing (Equation 5), substituting the wing dimensions with those of the vertical and horizontal tails. This approach can also be extended to supersonic conditions yielding a values for the zero lift drag (at $M = 0.25$) and wave drag ($M = 0.8$), respectively.

$$C_{D0_{emp}} = 0.0015$$

$$C_{Demp_{wave}} < 0.0002$$

The results presented in the table below are intended for preliminary design and assessment purposes and should be refined using higher-fidelity analyses in subsequent studies. The Roskam methodology provides data at discrete Mach numbers; therefore, interpolation was used in the graphs to estimate values across the remaining Mach range.

Table 7-5. Zero-lift drag coefficient components at varying Mach numbers

Mach (M)	$C_{D_{0wing}}$	$C_{D_{0fuselage}}$	$C_{D_{0emp}}$	$C_{D_{0wave}}$	$C_{D_{oTOTAL}}$
0.25	0.00520	0.02076	0.0015	0.000	0.02746
0.6	0.01480	0.02070	0.0026	0.001	0.03910
0.80	0.01483	0.02060	0.0028	0.006	0.04423
0.90	0.01509	0.02056	0.0031	0.0031	0.05175

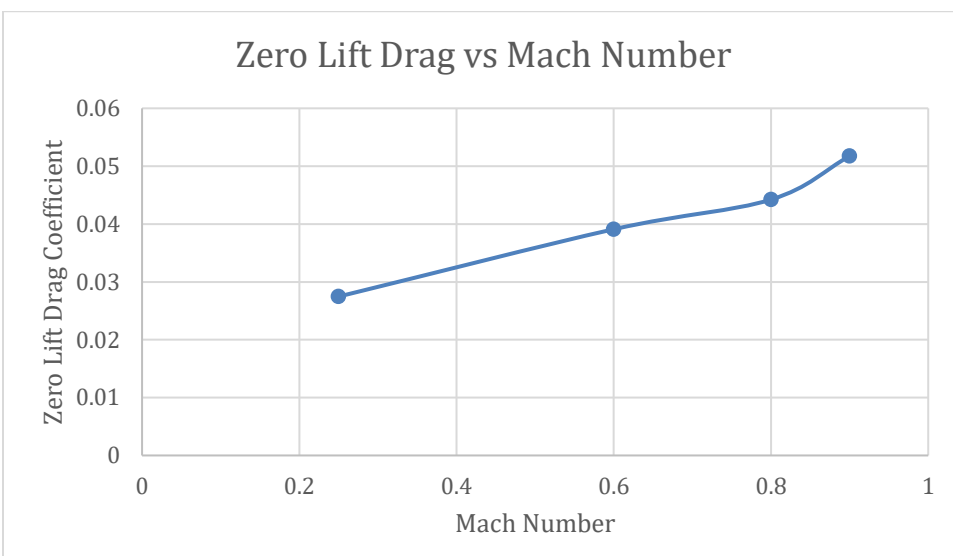


Figure 7-9. Zero Lift Drag vs. Mach Number

We now return to our original drag equation, incorporating the elements of the induced drag factor and zero lift drag to construct the drag polars.

Table 7-6. Drag Polar at specified Mach numbers

Mach Number (M)	Drag Polar (C_D)
0.25	$0.02746 + 0.1036{C_L}^2$
0.6	$0.03910 + 0.10756{C_L}^2$
0.8	$0.00423 + 0.1110{C_L}^2$
0.9	$0.05175 + 0.11310{C_L}^2$

Using the drag polars, we can construct relations between lift, drag, angle of attack, and Mach number.

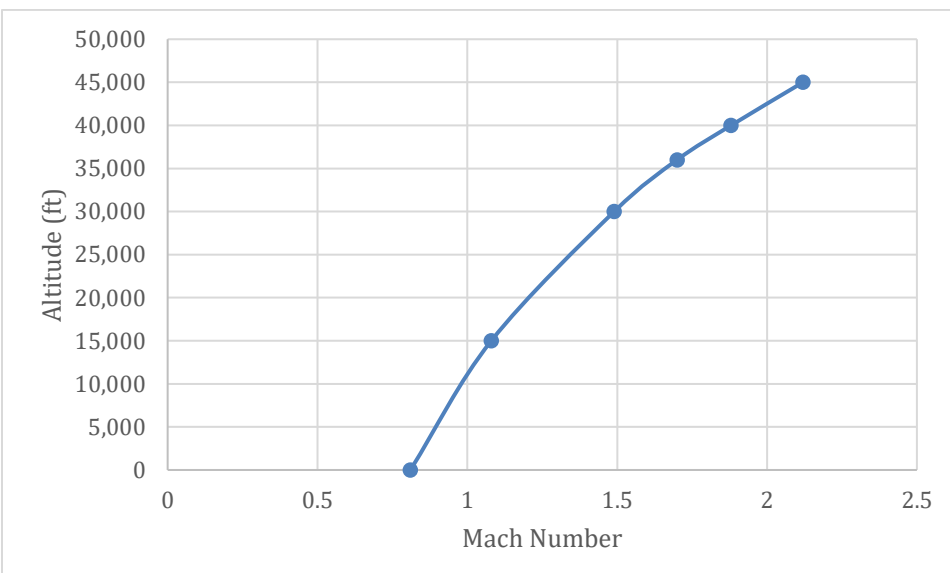


Figure 7-10. Altitude vs. Mach Number

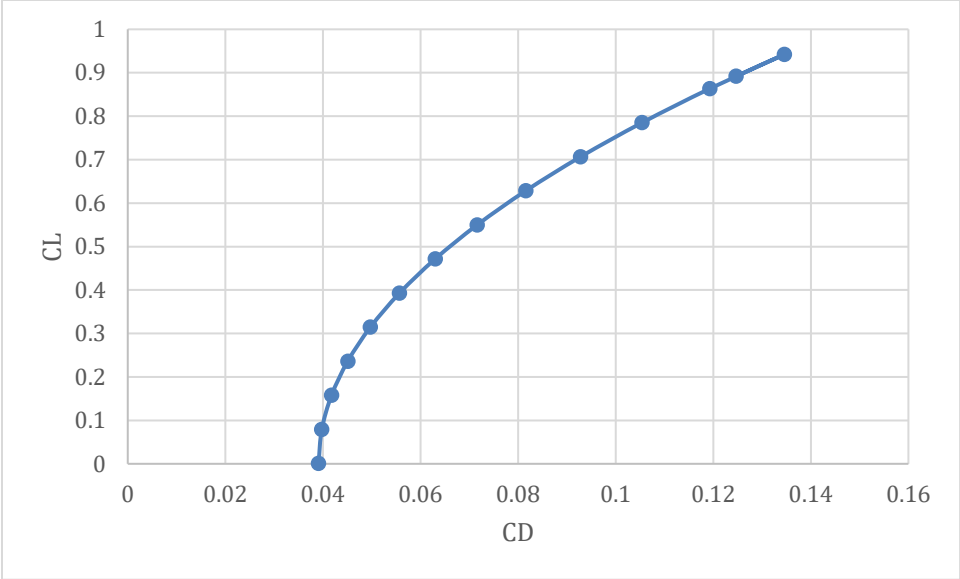


Figure 7-11. Coefficient of Lift vs. Coefficient of Drag

Table 7-7. Lift and Drag Coefficients and their ratio

Coefficient of Drag (C_D)	**Coefficient of Lift (C_L)**	C_L/C_D
0.0398	0.0008	0.02
0.039	0.079	1.99
0.042	0.158	3.77
0.045	0.236	5.24
0.050	0.315	6.32
0.055	0.393	7.05
0.063	0.471	7.50
0.072	0.550	7.70
0.082	0.628	7.70
0.093	0.707	7.60
0.105	0.785	7.44
0.119	0.864	7.24
0.135	0.942	7
0.125	0.89	7.15

The aircraft has a maximum lift to drag ratio $(L/D)_{max} = 7.70$.

To calculate the pitching moment of the aircraft, Roskam's method was once again utilized.

$$C_{m_{o_W}} = \left(A cos\Lambda_{c/4}\right)/\left(A + 2cos\Lambda_{c/4}\right)\frac{\left(c_{m_{o_r}} + c_{m_{o_t}}\right)}{2} + \left(\frac{\Delta C_{m_o}}{\varepsilon_t}\right)\varepsilon_t$$

$$C_{m_{o_W}} = \left(A cos(10.4°)\right)/\left(4 + 2cos(10.4°)\right)(-0.031) + (-0.001)(-0.0436)$$

$$C_{m_{o_W}} = -0.0204$$

We then calculate the lift curve slope and static margin:

$$\frac{dC_m}{dC_L} = \frac{x_{ac} - x_{cg}}{c}$$

$$\frac{dC_m}{dC_L} = Hn = 0.0041$$

The static margin is positive implying an instability in the aircraft. In order to address such instability, future iterations of the aircraft will incorporate a closed loop control system or a more forward center of gravity. The results are shown in the figure below.

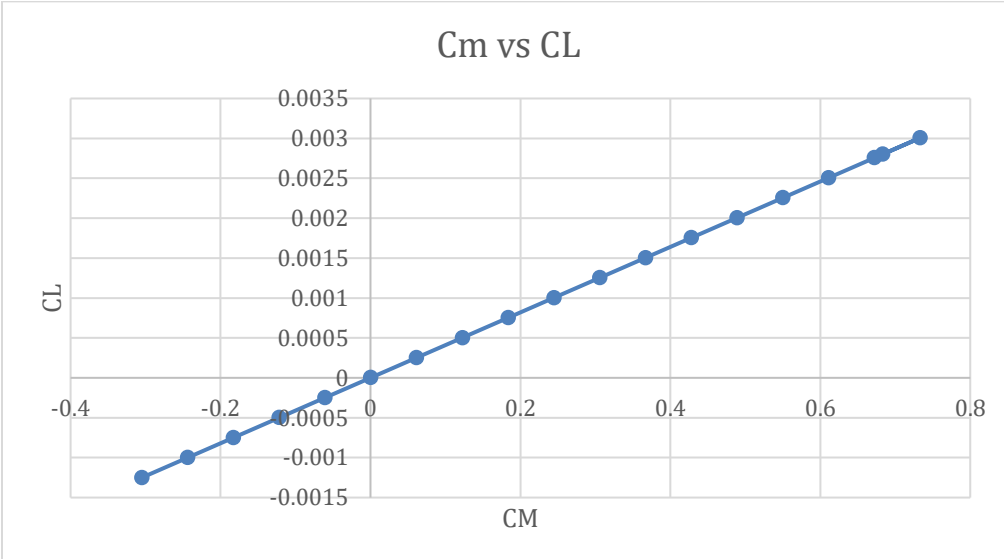


Figure 7-12. Lift coefficient vs. Moment coefficient

8 Propulsion

8.1 Engine Selection

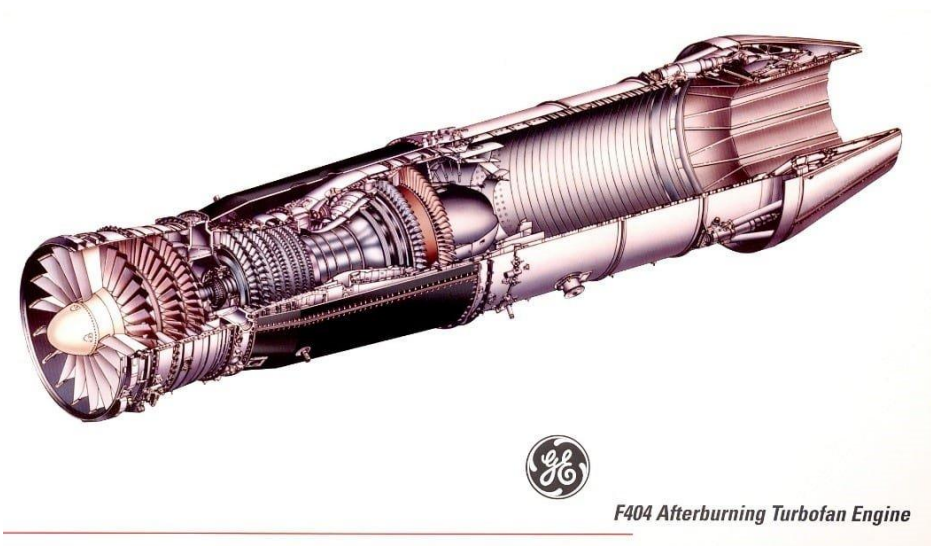


Figure 8-1: F404-GE-402 Afterburning Turbofan Engine

Table 8-1: F404-GE-402 Engine specifications

Engine Specification	Value
Dry / Afterburn Thrust	11,000 / 17700 lbf
Length	154 in
Maximum Diameter / Inlet Diameter	35 / 31 in
Airflow	146 lb/sec
Pressure Ratio	26:1
Engine Weight	2,282 lb (Per Engine)
TSFC (Dry / Afterburn)	0.80 / 1.90 lb/lbf*hr
Bypass Ratio (BPR)	0.34
Efficiency Coefficient	0.20 – 0.25

8.2 Inlet Sizing

Calculating the minimum inlet diameter required to provide sufficient airflow to the F404-GE-402 engines

$$\dot{m} = \rho V A$$

Equation 8-1

$$A_{min} = \frac{\dot{m}}{\rho V}$$

Equation 8-2

To compute the minimum inlet capture area, we are evaluating at a high-speed design point. In this case for our aircraft, the design Mach number is 0.9, sea level air density is 0.0765 $\frac{lb}{ft^3}$, and the air speed at sea level is assumed to be 1116.45 ft/s. From these parameters, freestream velocity can be calculated.

$$V = Ma$$

Equation 8-3

$$V = 0.9\left(1116.45\frac{ft}{s}\right) = 1004.81\frac{ft}{s}$$

With freestream velocity calculated, the minimum cross-sectional area for the aircraft inlet can now be acquired.

$$A_{min} = \frac{146\ lb/sec}{\left(0.0765\ \frac{lb}{ft^3}\right)\left(1004.8\frac{ft}{s}\right)} = 1.899\ ft^2$$

$$A_{min} = 273.47\ in^2$$

The actual inlet geometry is a rectangle with a semicircle outer edge. This geometry influenced fuselage design and was sized accordingly. For simplicity and comparison. The cross-sectional area can be converted to a diameter.

$$D_{min} = \sqrt{\frac{4A_{min}}{\pi}}$$

Equation 8-4

$$D_{min} = \sqrt{\frac{4(273.47\ in^2)}{\pi}} = 18.66\ in$$

The aircraft's actual inlet cross sectional area is $675.17\ in^2$. This inlet sizing was influenced by the fuselage geometry and was largely over sized to reduce inlet velocity. The reduced inlet velocity helps reduce the total pressure loss through the diffuser. This means more net thrust is available. The large inlet also helps improve afterburner stability and adequate airflow at high angles of attack. Lastly, this oversized inlet cross section allows for a future growth margin regarding the duct length, curvature, and convergence of airflow into the engine intake.

$$D_{\ max} = \sqrt{\frac{4A_{max}}{\pi}}$$

Equation 8-5

$$D_{max} = \sqrt{\frac{4(675.17)}{\pi}} = 29.32\ in$$

For comparison, the minimum and actual cross-sectional area and diameter are then computed as ratios of each other to showcase a margin for error coefficient.

$$\frac{A_{actual}}{A_{min}} = \frac{675.17\ in^2}{273.47\ in^2} = 2.47$$

Equation 8-6

$$\frac{D_{actual}}{D_{min}} = \frac{29.32\ in}{18.66\ in} = 1.57$$

Equation 8-7

This means the aircraft's current inlet configuration is 2.47 times larger than the minimum required inlet size. If the inlet is simplified to a diameter, the aircrafts current inlet is 1.57 times larger than the minimum required diameter. This means our inlet is properly sized for robustness and survivability of the aircraft. The inlets are designed to minimize ingestion of the boundary layer airflow past the nose cone. The inlets are to be mounted slightly off the fuselage to ensure high momentum free stream air makes it to the engine intake. Each F404-GE-402 is equipped with a convergent-divergent afterburning nozzle. These nozzles are sized and positioned to maximize high-temperature exhaust flow, provide the required thrust at sea level, and minimize the external drag after of the fuselage. The twin engine arrangement places the nacelles close to the aircraft centerline. This helps minimize the yawing moment if engine failure occurs.

The total afterburning thrust of 35,400 lbf for both engines provides substantial thrust. Our maximum takeoff weight is 42,567.47 lb. This gives us a thrust to weight ratio of 0.831. This thrust to weight ratio is sufficient for high climb rates, and sustained turn capability.

8.3 Propulsion Integration Design (In Progress)

8.4 Performance Data (In Progress)

9 Performance (In Progress)

9.2 Range

To calculate the total combat range of the F-44 Falcon, some design points and assumptions must be made.

Assumptions

1. TSFC – Because the thrust specific fuel consumption curves are not publicly available, a conservative estimate of 0.8 has been chosen.
2. Wi/Wf – For combat radius, a cruise velocity is assumed. Knowing that the segment of flight is a steady cruise, a Wi/Wf is assumed to be 0.73. Changes in this number cause the range to vary significantly when the value strays too far from 0.73. So in light of this sensitivity, 0.73 is assumed for the time being.

The calculation done for the combat radius is show using the Breguet Range equation in equation 9-1.

$$R = \frac{V}{TSFC}\left(\frac{L}{D}\right)\ln\left(\frac{Wi}{Wf}\right)$$

Equation 9-1

Table 9-1. Combat radius information

Combat Radius				
TSFC	**L/D**	**WI/WF**	**V (Knots)**	**Range (Nm)**
0.8	7.7	0.73	530	1605

9.5 Takeoff Analysis

The following performance items should be treated in the indicated sequence. For each, present a description of the procedure, a sample calculation, and the required graphs. Only need to present the procedure and sample calculation one time.

This section presents the carrier takeoff performance analysis for the proposed AIAA Navy strike fighter aircraft. The analysis is conducted in accordance with the AIAA Request for Proposal and the AE 460A Report Content Requirements. Only carrier takeoffs are considered in this section, as shore-based operations are not primary design drivers for the aircraft.

Takeoff performance is evaluated for the heaviest mission configuration (strike) as this case produces the highest catapult velocity required. The aircraft is to take off using the C13-2 catapult at sea-level tropical-day conditions (89.8°F) with zero wind over the deck, as specified in the AIAA RFP.

Assumed Value and Constants

Table 9-2. Assumed Values and Takeoff Conditions

Parameter	**Symbol**	**Value**	**Units**
Aircraft weight	W	42567	lbf
Aircraft mass	M	1323.025	slug
Air density at sea level (89.8°F)	ρ	0.00224	$slug/ft^3$
Wing reference area	S	600	ft^2
Wingspan	b	41.5	ft

Aspect ratio	AR	2.87	N/A
Oswald efficiency factor	e	0.768	N/A
Zero-lift drag coefficient	C_{DO}	0.045	N/A
Lift coefficient (takeoff)	C_L	0.9	N/A
Drag coefficient	C_D	0.1459	N/A
Lift-induced drag factor	k	0.125	N/A
Rolling friction coefficient	μ	0.02	N/A
Available thrust (one engine)	T	35400	lbf
Distance down runway	x	0-300	ft

To find the takeoff speed required for aircraft design we need to find the value at which the aircraft stalls first. This is done using the equation:

$$V_{Stall\,Takeoff} = \sqrt{\frac{2W}{\rho S C_{L\,max}}}$$

Equation 9-2

$$V_{stall,Takeoff} = \sqrt{\frac{2(42567)}{(0.00224)(600)(0.9)}} = 257.4831\ \left(\frac{ft}{s}\right)$$

Once we have found the stall velocity, we can find what the necessary takeoff velocity is using Equation 9-3.

$$V_{Takeoff} = 1.2V_{Stall\,Takeoff}$$

Equation 9-3

$$V_{Takeoff} = 1.2(257.4831) = 308.9797\ \left(\frac{ft}{s}\right)$$

From this value we then can calculate the acceleration needed to reach takeoff velocity over the length of the deck. This is done by taking the takeoff velocity squared over two times the runway distance as shown:

$$a_{Takeoff} = \frac{{V_{Takeoff}}^2}{2(x)}$$

Equation 9-4

$$a_{Takeoff} = \frac{308.9797^2}{2(300)} = 159.1141\ (\frac{ft}{s^2})$$

Solving for the acceleration required to reach takeoff velocity over the span of the runway we can now get how much force the catapult will need to produce in tandem with the engines to reach the required takeoff speed. This is done through subtracting the acceleration that is produced from the thrust of the engines from the acceleration required for takeoff to get first the catapult acceleration as shown:

$$a_{Catapult} = a_{Takeoff} - \frac{T}{M}$$

Equation 9-5

$$a_{Catapult} = 159.1141 - \frac{35{,}400}{1323.025} = 132.3572\ (\frac{ft}{S^2})$$

Now with the catapult acceleration we can get the force the catapult must produce to achieve takeoff speed. This is done by multiplying the acceleration of the catapult by the mass of the aircraft.

$$F_{Catapult} = a_{Catapult}M$$

Equation 9-6

$$F_{Catapult} = (132.3572)(1323.025) = 175{,}112\ (lbf)$$

With the force of the catapult, we can find the net force acting on the plane in the horizontal direction during takeoff through the equation:

$$F_{Net\ i} = T + F_{Catapult} - F_{FricTakeoff\ i} - D_{Takeoff\ i}$$

Equation 9-7

Solving Equation 9-7 for the instantaneous net force at the start of the runway (x=0)

$$F_{Net\ i} = 35{,}400 + 175{,}112 - 851.34 - 0 = 209{,}661\ (lbf)$$

Where

$$F_{FricTakeoff\ i} = \mu(W - L_i)$$

Equation 9-8

Evaluate Equation 9-7 at runway start (x=0)

$$F_{FricTakeoff\ i} = 0.02(42576 - 0)$$

With

$$L_i = q_i S C_L$$

Equation 9-9

Evaluate Equation 9-8 at runway start (x=0)

$$L_i = (0)(600)(0.9) = 0$$

And

$$D_{Takeoff\ i} = q_i S C_D$$

Equation 9-10

Evaluate Equation 9-9 at runway start (x=0)

$$D_{Takeoff\ i} = (0)(600)(0.145925) = 0$$

With the value of q being solved instantaneously for each velocity value as you go down the length of the runway using:

$$q_i = \frac{1}{2}\rho V^2$$

Equation 9-11

Evaluate Equation 9-10 at runway start (x=0)

$$q_i = \frac{1}{2}(0.00224)(0^2)$$

Achieving the values for the runway start, we then can solve for instantaneous velocity based on runway position from our net force being converted to an acceleration and using the previous velocity:

$$V_{i+1} = \sqrt{V^2{}_i + 2(\frac{F_{net\ i}}{M})(x_{i+1} - x_i)}$$

Equation 9-12

Evaluating Equation 9-11 at a value on the runway (x=5) with the initial value being at runway start (x=0).

$$V_{i+1} = \sqrt{0 + 2\left(\frac{209{,}661}{1323.025}\right)(5-0)} = 39.8083\ \left(\frac{ft}{s}\right)$$

For the example we used an increment of five feet down the runway, this can be done at higher or lower increments depending on the accuracy you want to convey. With the new value for instantaneous velocity, we can plug in to Equation 9-10 to get the new dynamic pressure at your instantaneous position. Then using the new dynamic pressure, you can plug into Equation 9-9 and Equation 9-8 to get the instantaneous lift and drag at your position. Using the lift to solve Equation 9-7 you get the instantaneous friction force. With all these values you're able to then solve Equation 9-6 to get the instantaneous net force. To continue this process, you then solve equation 9-11 again and repeat. By doing this process

we can see the building of velocity, lift and drag down the distance of the runway during takeoff.

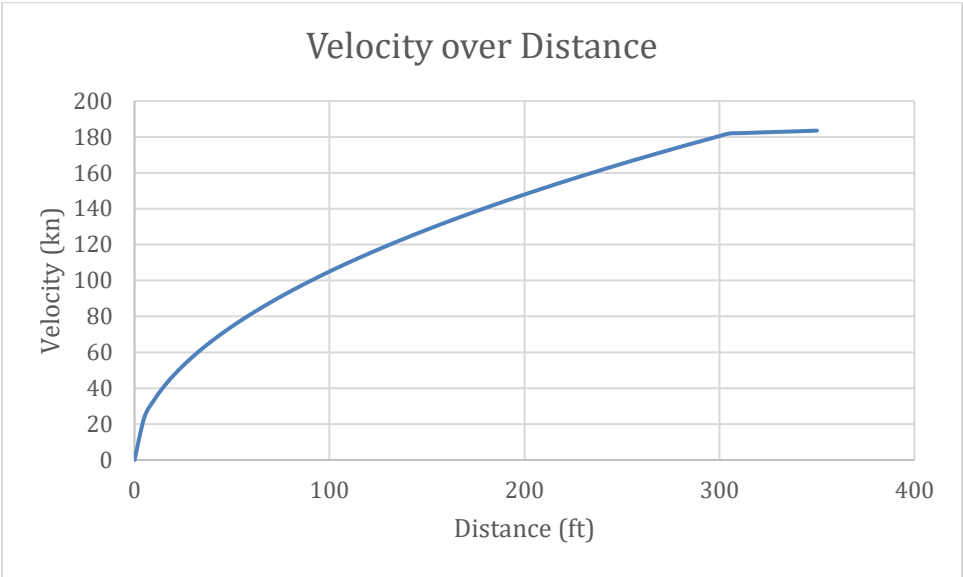


Figure 9-1. Takeoff Velocity Over Distance

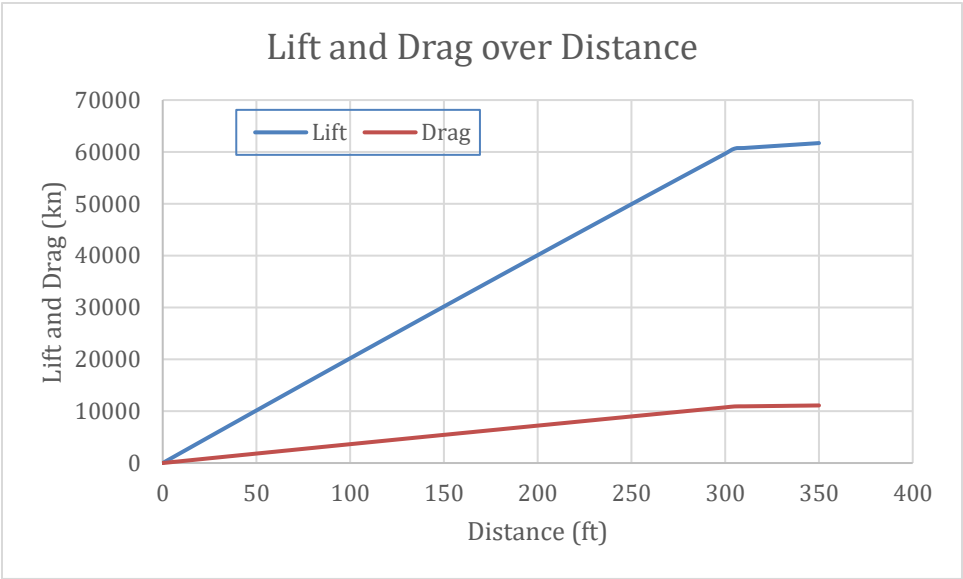


Figure 9-2. Takeoff Lift and Drag Over Distance

In both Figure 9-1 and Figure 9-2 we can see that as the departure of the aircraft from the catapult and runway occurs the slopes decrease drastically. This is resulting from the aircraft no longer having both catapult and engines combining to produce force.

Table 9-2 Takeoff Calculated Values

Parameter	Value	Units

$V_{Stall\ Takeoff}$	152.43	knots
$V_{Takeoff}$	182.92	knots
$a_{Takeoff}$	94.19	knots/s
$a_{Catapult}$	78.36	knots/s
$F_{Catapult}$	175,112	lbf

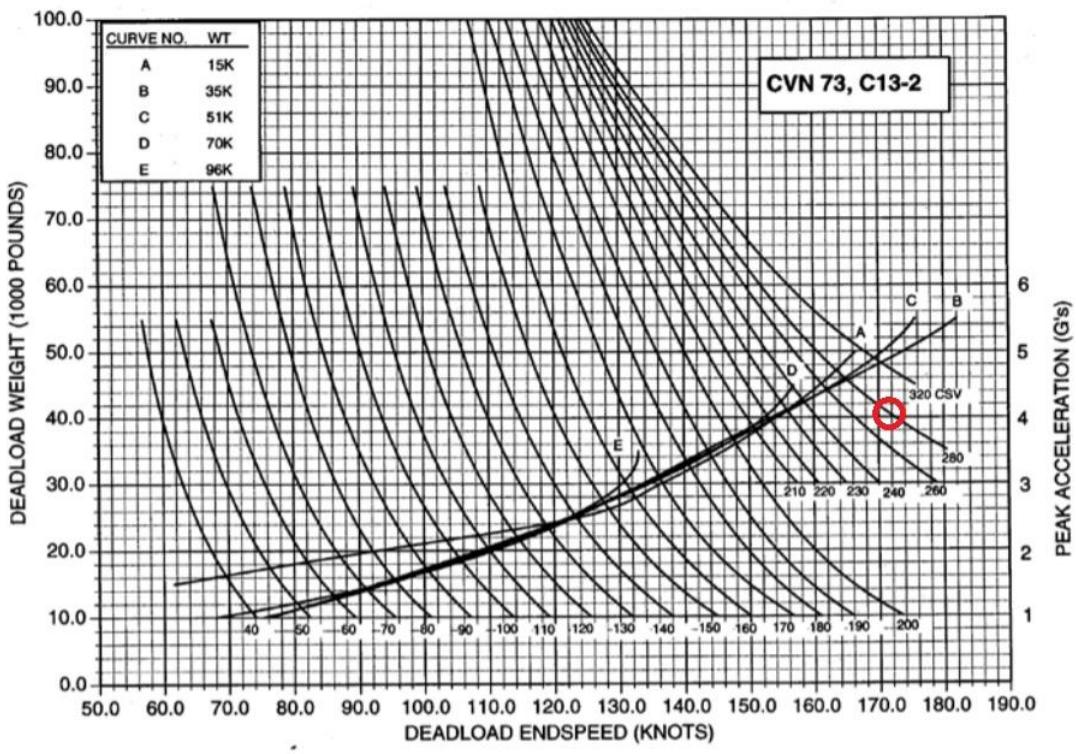


Figure 9-3. Catapult Parameters

Taking the calculated takeoff velocity and applying it to the chart for catapult capabilities in Figure 9-3 it looks like we are currently unable to hit the correct takeoff speed. However, this is because we have assessed for a takeoff speed at a factor of one point two times the stall speed. This is not always the required speed for takeoff but a safe value to allow for mishaps during takeoff performance. The relationship between lift and weight is the true deciding factor as to when takeoff is achievable. We can see that lift equals weight at roughly 215 feet down the runway through our iteration process. At this point the aircraft speed is equal to 153.2849 (kn). We start to see lift become greater than aircraft weight as we continue down the runway ending with a lift value at the full length of the runway of 59664.5 (lbf) and velocity of 180.4645 (kn). This means we will

be able to fit onto the capabilities of the catapult with our end stroke speed at the maximum possible for our aircraft weight. This area is circled in Figure 9-3 at about 173 (kn).

Takeoff is possible under the current parameters, but it is not satisfactory. We plan to have an increase in our lift coefficient which will make takeoff velocity calculated from the stall velocity a lower value. This will make it easier for the aircraft to fit into the catapult parameters when we see increases in weight or any other changes that may be necessary.

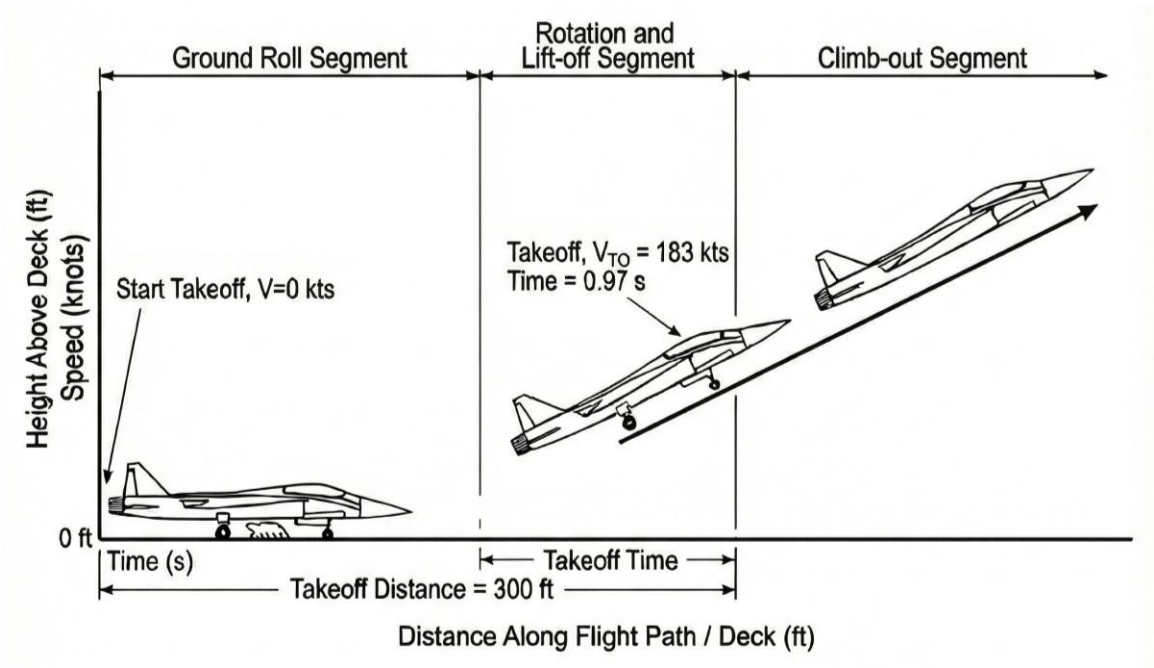


Figure 9-4. Takeoff Profile

The AIAA parameters for takeoff ask for the highest velocity for the minimum catapult end stroke speed of four different scenarios. After evaluation of the four parameters, we found that we will be between a velocity value at 90 percent of our max lift coefficient value or we will be at the minimum velocity to drop no more than 10 feet off the deck after 32 feet of deck run. We need to solidify our parameter for velocity in the future by solving for the velocity that gives us the closest to 10 feet of drop possible and comparing it to our 90 percent of lift coefficient value.

9.6 Landing Analysis

This section presents the carrier landing performance analysis for the proposed AIAA Navy strike fighter aircraft. The analysis is conducted in accordance with the AIAA Request for Proposal and the AE 460A Report Content Requirements. Only carrier

arrested landings are considered in this section, as shore-based operations are not a primary design driver for the aircraft.

Landing performance is evaluated for the heaviest mission configuration (strike) as this case produces the most conservative approach speeds and arrestment loads. The aircraft is assumed to land in the Weapons Landing (WL) configuration, with landing gear down, full landing flaps deployed, speed brakes available after touchdown, and tailhook down.

All calculations assume sea-level tropical-day conditions (89.8°F) with zero wind over the deck, as specified in the AIAA RFP.

Assumptions and constants

Table 9-3. Assumed Values and Landing Conditions

Parameter	**Symbol**	**Value**	**Units**
Wing reference area	S	600	ft^2
Maximum landing configuration lift coefficient	$C_{L,max,L}$	0.9	N/A
Air density at sea level (89.8°F)	ρ	0.00224	$slug/\,ft^3$

General equations

Landing stall speed:

$$V_{stall,L} = \sqrt{\frac{2W}{\rho\, S\, C_{L,max,L}}}$$

Equation 9-13

Approach speed:

$$V_{app} = 1.1 V_{stall,L}$$

Equation 9-14

Engaging speed:

$$V_{eng} = 1.05 V_{app} = 1.155 V_{stall,L}$$

Equation 9-15

Arrestment Landing Weights

Landing performance was evaluated at three representative weights for the Strike mission in Weapons Landing (WL) configuration to capture the effect of fuel state on carrier recovery performance. The maximum landing weight corresponds to the Strike configuration with full internal fuel and stores and represents the upper bound of the full-

fuel to zero-fuel sweep and is included for sensitivity only. The arrestment landing weight is the primary design case and is defined per the AIAA RFP, including fuel for loiter and landing attempts, 25% of maximum fuel weight, and 50% of store weight, and represents the expected operational carrier recovery condition. The minimum landing weight corresponds to the aircraft with zero usable fuel while retaining 50% of the Strike store weight in the WL configuration and defines the lower bound of the landing weight range.

Table 9-4. Landing Weights

Description	**Weight, W (lb)**
Max WL Weight (full fuel Strike)	42,567
Arrestment Landing Weight (design case)	29,538
Min WL Weight (0 fuel, 50% stores)	22,205

AIAA Arrestment Landing Weight (design landing case):

$$W_{ALW} = W_{empty} + W_{loiter+attempts} + 0.25W_{fuel,max} + 0.50W_{stores}$$

Equation 9-16

Where:

- W_{empty}= empty configuration weight
- $W_{loiter+attempts}$= fuel required for 20 min loiter at 10,000 ft & two landing attempts
- $W_{fuel,\max}$ = maximum fuel weight for the Strike mission
- W_{stores}= Strike stores weight

Maximum WL weight:

$$W_{max} = W_{full-config\ Strike}$$

Equation 9-17

Minimum WL weight:

$$W_{min} = W_{empty} + W_{trap} + 0.50W_{stores}$$

Equation 9-18

Landing Speeds

Corresponding landing velocities were calculated for each of the three defined weight cases to establish the operational envelope for carrier recovery. The stall speed was determined as a function of the aircraft's weight and the current maximum landing lift

coefficient of 0.9. In accordance with the design requirements, the approach speed includes a 10% safety margin above stall, while the arrestment engaging speed is calculated as 5% above the approach speed. The resulting speeds for the Minimum, Design, and Maximum landing weights are summarized below.

Table 9-5. Landing Speeds

Condition	**Weight (lbs)**	V_{app} **(knots)**	V_{eng} **(knots)**
Min Landing	22,205	124.87	131.12
Design Landing	29,538	144.03	151.23
Max Landing	42,567	172.90	181.55

9.6.1

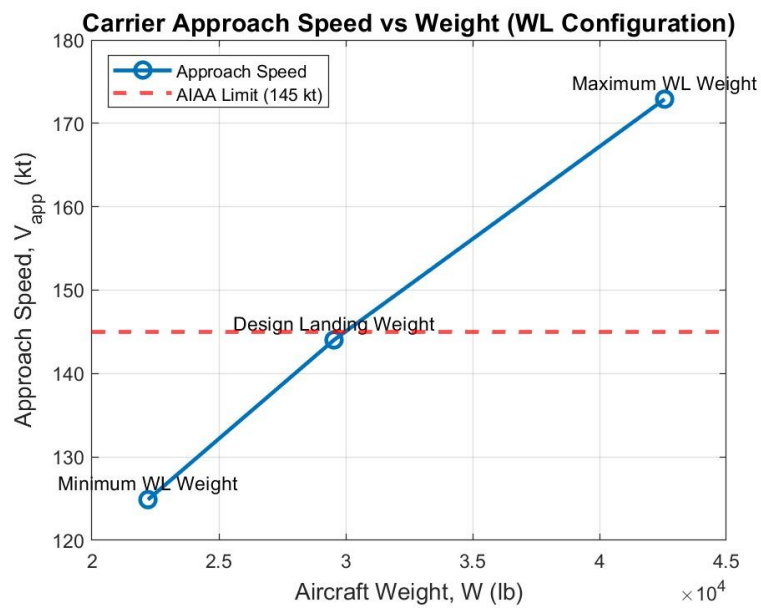


Figure 9-5. Approach Speed vs. Weight

Sample Calculations for Designed Landing Weight

Landing stall speed:

$$V_{stall,L} = \sqrt{\frac{2(29538)}{(0.00224)(600)(0.9)}} = 221.0\frac{ft}{s} = 130.94\ knots$$

Approach speed:

$$V_{app} = 1.1(130.94) = 144.03\ knots$$

Engaging speed:

$$V_{eng} = 1.05(144.03) = 151.23\ knots$$

Arresting Gear Performance

The calculated engaging speed of 151.23 knots currently exceeds the maximum operational speed limit of the Mark 7 Mod 3 arresting gear, as indicated by the vertical limit line in Figure 9-5. This over-speed condition is driven by the conservative lift coefficient of 0.9 used for this preliminary analysis. As detailed in the Landing Lift Capability section below, increasing the $C_{L,max,L}$ to a target of 1.2 or greater in the next design iteration will reduce the approach velocity below 145 knots, bringing the aircraft fully within the safe recovery envelope of the arresting gear.

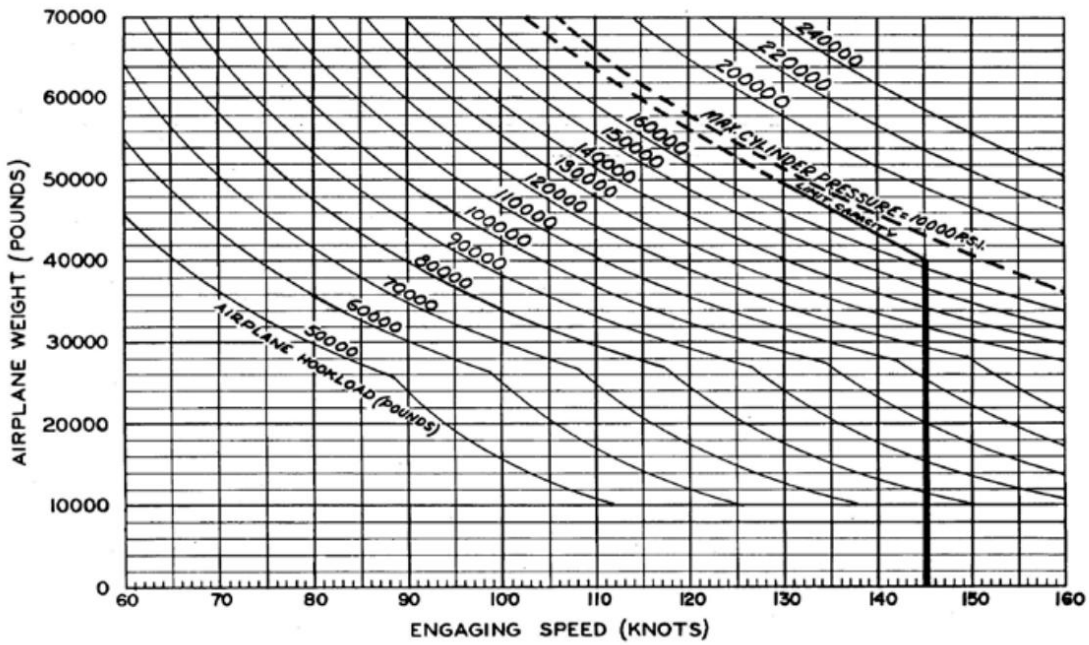


Figure 9-6. Mark 7 Mod 3 Arresting Gear Performance Curves

Landing Lift Capability

A landing lift coefficient of 0.9 was used for this analysis. This value is conservative and does not represent a final carrier-capable design. Carrier-based fighter aircraft typically operate with landing lift coefficients between 1.8 and 2.2.

At the current arrestment landing weight, a minimum landing lift coefficient of about 1.2 is required to meet the AIAA approach speed limit of 145 knots. With the current value of 0.9, approach speeds exceed this limit and do not meet the requirement.

For this reason, a target landing lift coefficient of approximately 2.0 is planned for later design iterations to provide sufficient margin for carrier recovery.

Planned Design Improvements

In later design phases, the landing lift capability will be improved by updating the high-lift system. This includes increasing trailing-edge flap deflection, implementing leading-edge devices, and evaluating drooped ailerons in the landing configuration.

Higher-fidelity aerodynamic methods will be used to better estimate landing lift performance. Approach speed, engaging speed, and arrestment loads will then be recalculated using the updated lift capability.

Landing Profile

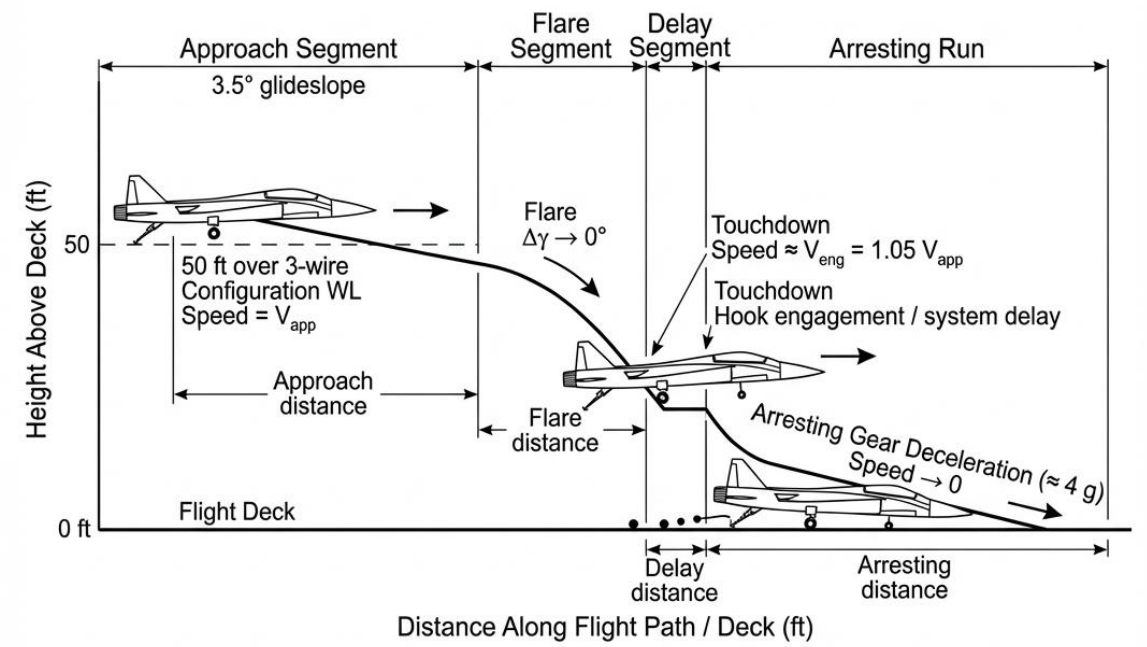


Figure 9-7. Landing Profile

One Engine Inoperative (OEI)

For calculating a Reynolds number at 100 knots true air speed we plug in the assumption for mean aerodynamic chord for the equation:

$$Re = \frac{V(MAC)}{\mu}$$

Equation 9-19

Where μ is the kinematic viscosity of 90 degrees Fahrenheit at sea level

$$Re = \frac{168.781(16.5998)}{(1.68 \times 10^{-5})} = 1.6677 \times 10^{8}$$

For the takeoff weight in our one engine inoperative calculations, we use:

$$W_{TO} = W_{Ramp,Max} - W_{f,burned}$$

Equation 9-20

With $W_{f,burned}$ being the weight of fuel burned in 5 minutes of idle time and the $W_{Ramp,Max}$ is 1.01 times the max takeoff weight. The fuel is burned at 9,000 lbf/hr making $W_{f,burned} = 750\ (lbf)$. We will use the value for the strike mission of 42567 (lbf). This will give us a $W_{Ramp,Max} = 42{,}992.67\ (lbf)$. Plug these values in and we get:

$$W_{TO} = 42{,}992.67 - 750 = 42{,}242.67\ (lbf)$$

9.7.3 Critical Field Length

The critical field length analysis for the one engine inoperative takeoff condition is currently in progress. The governing equations for accelerating with takeoff and accelerating with stop performance have been established based on the force balance during the takeoff roll, accounting for thrust, drag, rolling friction, and braking forces. The methodology employs numerical integration of acceleration and deceleration to determine runway distance as a function of engine failure. Results and a field length determination will be completed following full evaluation of the acceleration and braking segments.

9.7.4 Velocity Dependent Variation of Critical Field Length

The velocity dependent variation of critical field length is under development. This analysis will be performed by evaluating the acceleration with takeoff and accelerating with stop distances over a range of decision speeds and identifying the required runway length at each speed. The resulting critical field length curve will be found once the remaining integrations are completed.

9.7.5 OEI Rate of Climb vs. Weight (Gear Down)

The one engine inoperative rate of climb for the wing landing configuration with landing gear down is in progress. The analysis has been defined using excess thrust and drag relationships to evaluate climb as a function of aircraft weight. Multiple weight conditions with spanning fuel values will be analyzed to find OEI climb performance. Results and corresponding rate of climb versus weight plot will be done upon completion of the remaining calculations.

9.7.6 OEI Rate of Climb vs. Weight (Gear Up)

The one engine inoperative rate of climb analysis for the wing landing configuration with landing gear up is in progress. The methodology mirrors the gear down analysis, with

updated aerodynamic drag characteristics except with the retracted landing gear configuration. Rate of climb performance will be evaluated across multiple aircraft weights to show the OEI climb capability compared to the gear down case. Final results and graphical presentation will be done once the performance calculations are finalized.

9.8 Performance Summary

Flight Envelope

The Operating Flight Envelope was calculated at a gross weight of 39,943 lbs, corresponding to the 60% internal fuel condition. This plot illustrates the aircraft's performance boundaries:

Aerodynamic Stall Limit (Left Boundary): Defines the minimum flyable speed based on the maximum lift coefficient. As altitude increases, the true airspeed required to sustain lift rises.

Thrust Limit (Right Boundary): Represents the maximum level-flight speed where drag equals available thrust. Notably, the current analysis indicates a top speed below Mach 1. This operational ceiling appears incorrect given the aircraft's design requirement for a Mach 1.6 dash speed, suggesting the preliminary drag model may be overestimating transonic wave drag.

Absolute Ceiling: The peak of the envelope occurs where the stall and thrust limits converge. Above this altitude, the aircraft cannot maintain steady level flight.

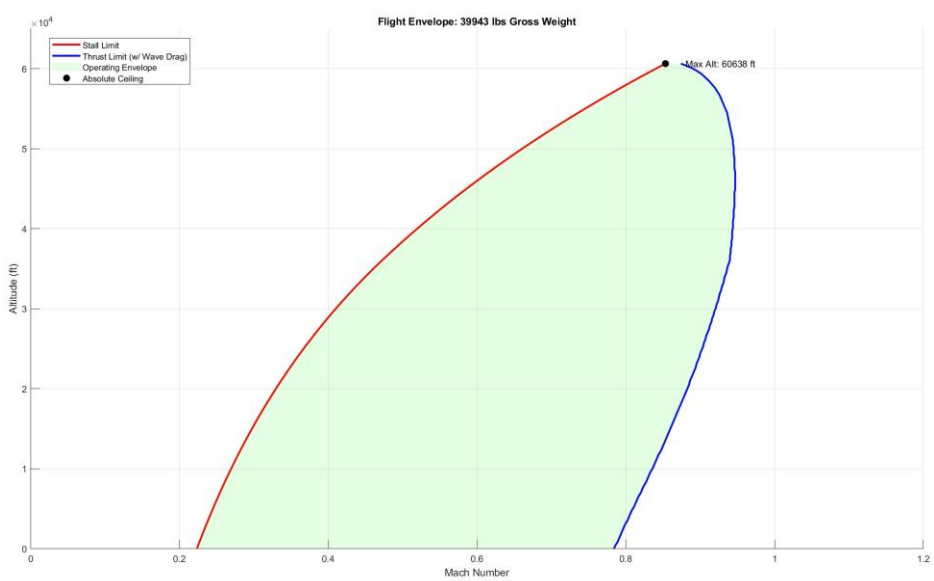


Figure 9-8: Operating Flight Envelope

Performance Summary

Table 9-6: Performance Summary

Paragraph	Requirement Description	Required Value	Achieved Value
3.2.1.a	Carrier Launch	Capable	Capable
3.2.2.b	Carrier Recovery Approach Speed (V_app)	< 145 kts	144 kts
3.4.1.a	Combat Radius (Air-to-Air / Strike)	700 nmi	800 nmi
3.4.1.e	Dash Speed (30000 ft)	Mach 1.6	Mach 1.49
3.4.2.c	Sea Level Dash Speed	Mach 0.85	Mach 0.81

10 Stability and Control (In Progress)

An aircraft is a six–degree-of-freedom dynamic system, with three translational and three rotational motions. These motions are governed by three linear and three angular accelerations under gravitational, inertial, aerodynamic, and propulsive forces and moments. In steady flight, static equilibrium applies, requiring lift to balance weight and thrust to balance drag. Static longitudinal stability is achieved when a disturbance produces a restoring nose-down pitching moment that returns the aircraft to equilibrium and is done by using a horizontal stabilizer. The aircraft uses properly sized and positioned horizontal stabilizers to generate corrective aerodynamic forces, ensuring longitudinal stability and controlled pitch behavior throughout all phases of flight. With the addition of a closed-loop feedback system of a quadruplex fly-by-wire (FBW) system to avoid any pitch-up tendencies due to the center of gravity being located behind the aerodynamic center; this also allows the aircraft to be highly maneuverable.

Lateral and directional degrees of freedom are inherently coupled, as sideslip induces rolling and yawing effects and yawing motion induces both rolling motions and sideslip. Static directional stability refers to the aircraft's ability to naturally realign with the relative wind and eliminate sideslip caused by disturbances. While such disturbances may alter the aircraft's orientation, its overall heading should remain unchanged. Directional stability is governed primarily by sideslip and yaw behavior. To achieve this, the aircraft is designed with dual vertical stabilizers, which generate corrective aerodynamic forces to oppose unwanted yaw. Their size and aerodynamic efficiency are designed to provide

stable, predictable yaw control under varying flight conditions. Table 10-1 lists the empennage characteristics of the aircraft.

Table 10-1: Empennage sizing

Parameter	Units	Value
Horizontal Tail Area	ft^2	90
Horizontal Tail Span	ft	19
Taper Ratio (Horizontal)	-	0.4
LE Sweep Angle (Horizontal)	deg	29
Aspect Ratio (Horizontal)	-	4.01
Horizontal Tail Coefficient	-	0.163
Vertical Tail Area	ft^2	101.77
Vertical Tail Span	ft	12
Taper Ratio (Vertical)	-	0.2
LE Sweep Angle (Vertical)	deg	37
Aspect Ratio (Vertical)	-	2.83
Vertical Tail Coefficient	-	0.073

11 Configuration Spreadsheet

TBD

12 Life Cycle Costs

The Life Cycle Cost of the aircraft was determined using an estimation technique from Nikolai. All the equations used in Nikolais techniques give a resulting cost in the dollar value from 1998. To correct this, all the values are multiplied by a factor of 1.99 to give a dollar value in the current year. The weight influencing the cost of the aircraft is the American Manufacturers Planning Report (AMPR Weight). This is the weight which does not include any instruments, wheels, electrical equipment, and engines. During the conceptual design phase, this is very hard to acquire because it takes detailed weight summaries. Typically, the AMPR weight is 62% of an aircraft's empty weight. For the cost methodology, the empty aircraft weight can be used with the 62% being accounted

for with coefficients. Table 12-1 shows the variables that are used in order to acquire the cost estimation.

Table 12-1: Nomenclature

R	Labor Rate
Y	Year
W	Empty Weight of Aircraft
S	Max Speed of Aircraft
Q	Quantity of Aircraft
E	Engineering Hours
D	Development Support Cost
F	Flight Test Operations Cost
T	Tooling Hours
L	Manufacturing Labor Hours
QC	Quality Control Hours
M	Material and Equipment Cost
P	Engine Cost
T_{SLS}	Maximum Sea Level Thrust (lbf)
M_{max}	Maximum Mach Number
T_R	Turbine Inlet Temperature (R)
A	Avionics Cost
W_A	Weight Avionics

The information in Table 12-2 is necessary in order to begin the calculations for the cost estimation. This is information that is produced from the design phase. For the labor rates we can extrapolate the values for the current year based on the increasing trend of hourly rate over the years. The following rates are found through:

$$R_E = 2.576Y - 5058$$

Equation 12-1

$$R_T = 2.883Y - 5666$$

Equation 12-2

$$R_M = 2.316Y - 4552$$

Equation 12-3

$$R_{QC} = 2.60Y - 5112$$

Equation 12-4

Plugging in the proper year will give you the rates shown. When solving for the developmental hours and costs, we will use the number of flight test aircraft from Table 12-2 as the quantity. To solve production, we will use the production quantity.

Table 12-2: DT&E Background Information

Time Frame of Costing	2025
TOGW	42567 (lb)
Empty Weight	19917 (lb)
Avionics Weight	2500 (lb)
Maximum Speed	1001 (kt)
Maximum Mach	2
Engine	F404-GE-402 T_{SLS} = 17,000 (lbf) T_R = 3,800 (R)
Flight Test Aircraft Number	9
Production Quantity for Costing	500
Labor Rates Engineering R_E Tooling R_T Manufacturing R_M Quality Control R_{QC}	For 2025 $158.40 per hour $172.08 per hour $137.90 per hour $153.00 per hour

To find the amount of engineering hours required for the development and production phases of the aircraft we use:

$$E = 4.86W^{0.777}S^{0.894}Q^{0.163}$$

Equation 12-5

Plugging in the information from table 12-2, we see the engineering hours required for the airframe. We multiply this value with our hourly rate obtained from Equation 12-1 to what this will cost. The production quantity is found by taking the total from aircraft test number and production quantity as our Q value for Equation 12-5, then subtracting the development hours.

$$E_P = (4.86W^{0.777}S^{0.894}(Q_D + Q_P)^{0.163}) - E_D$$

Equation 12-6

Similarly to finding the cost for development, we multiply the hours found for sustaining engineering in production by the engineering rate found in Table 12-2.

Table 12-3: Engineering Hours

Development	E_D = 7,329,305.97 hours
Cost	$1,160,962,065.24
Production	E_P = 14,148,692.94 - 7,329,305.97 E_P = 6,819,386.97 hours
Cost	$1,080,190,892

The development support cost is found through plugging in the values from Table 12-2 to:

$$D = 66W^{0.63}S^{1.3}$$

Equation 12-7

This will give us the cost in dollars from 1998 and upon applying the factor to convert to the current year we get the correct estimation.

Table 12-4: Development Support

Cost	$533,915,322

Similarly to development support, flight test operations also will give us a cost that will need to be converted to the current year using:

$$F = 1852W^{0.325}S^{0.822}Q^{0.263}$$

Equation 12-8

Plugging in the values from Table 12-2 and applying the conversion to the current year, we get the cost.

Table 12-5: Flight Test Operations

Cost	$384,354,598

When solving tooling we first find the hours required for development. This is done using:

$$T = 5.99W^{0.777}S^{0.696}Q^{0.263}$$

Equation 12-9

Plug the information from Table 12-2 and we get developmental hours. Multiply this by the rate for tooling and we get the cost. For the production cost of tooling, we do the same method for finding engineering production hours. Instead of utilizing Equation 12-6 however, we have a new one based on Equation 12-9:

$$T_P = (5.99W^{0.777}S^{0.696}(Q_D + Q_P)^{0.263}) - T_D$$

Equation 12-10

Plugging in the values from Table 12-2, we then can find the production hours required. Multiply this by the hourly rate and the cost associated is found.

Table 12-6 Tooling

Development	T_D = 2,865,442.57
Cost	$493,071,030
Production	T_P = 8,281,189.17 - 2,865,442.57 T_P = 5,415,746.60
Cost	$931,914,596

Manufacturing labor also follows the process for tooling and engineering, but with its own coefficients:

$$L = 7.37W^{0.82}S^{0.484}Q^{0.641}$$

Equation 12-11

Solve the development hours through Equation 12-11 from the values in Table 12-2, and you can then find the development cost associated as well as move on to solve the production values with:

$$L_P = (7.37W^{0.82}S^{0.484}(Q_D + Q_P)^{0.641}) - L_D$$

Equation 12-12

Using Table 12-2 and the value for development hours from Equation 12-11, we can find the hours required for production and the resulting price.

Table 12-7: Manufacturing Labor

Development	L_D = 2,862,270.32
Cost	$394,707,076

Production	L_P = 38,023,080.32 - 2,862,270.32 L_P = 35,160,810.00
Cost	$5,243,382,776

For quality control, we base it on a factor of the manufacturing labor hours required:

$$QC = 0.13L$$

Equation 12-13

Plugging in the values from Table 12-7 for both the development and production hours, we get the quality control hours. From these two values, we use Table 12-2 for the hourly rate of quality control and multiply for the respective prices shown in Table 12-8.

Table 12-8: Quality Control

Development	QC_D = 372,095.14
Cost	$56,930,556
Production	QC_P =4,570,905.30
Cost	$699,348,511

Like quality control, materials and equipment pricing are found using the same equation:

$$M = 16.39W^{0.921}S^{0.621}Q^{0.799}$$

Equation 12-14

The difference with material and equipment pricing is that for the quantity we plug in each of the respective quantities for development and production to get their costs. This cost then needs to be converted to the current year, and we are then given the values in Table 12-9.

Table 12-9: Material and Equipment

Development Cost	$125,523,948
Production Cost	$3,110,024,848

Finding the Engine cost requires us to first find the unit cost of each engine. This can be done using the equation:

$$P = 2306(0.043T_{SLS} + 243.3M_{max} + 0.969T_R)$$

Equation 12-15

From this equation, we find the unit cost of each engine. Then we multiply by the number of engines and the number of aircraft in development and production separately. This

will give the cost of 1998 dollars, which then gets converted to the current year. The results are shown in Table 12-10.

Table 12-10: Engine Cost

Engine	F404-GE-402 T_{SLS} = 17,000 (lbf) T_R = 3,800 (R)
Unit Cost	$12,260,729
Development	Assume 3 for Each Aircraft
Cost	$331,039,707
Production	Assume 2 for Each Aircraft
Cost	$12,260,729,892

For avionics costs, we base them off the weight of the avionics rather than the empty aircraft:

$$A = 16.39W_A^{0.921}S^{0.621}Q^{0.799}$$

Equation 12-16

We solve for development first and are given the cost in 1998 dollars. This value is then converted to the current year for an accurate estimation. After having the development cost before conversion, we can find the production cost using:

$$A_P = \left(16.39W_A^{0.921}S^{0.621}(Q_D + Q_P)^{0.799}\right) - A_D$$

Equations 12-17

Plugging in from Table 12-2, we get the production cost value of 1998 dollars. Convert to the current year and the results are shown in Table 12-11.

Table 12-11: Avionics

Development Cost	$18,562,826
Production Cost	$447,958,872

After finding all of the developmental costs we can sum them all together to get our final subtotal for the developmental, test and evaluation of the project. This can be seen in Table 12-12.

Table 12-12: Total DT&E Cost

Airframe Engineering	$1,160,962,065.24

Development Support	$533,915,322
Flight Test Aircraft	$384,354,598
Tooling	$493,071,030
Manufacturing Labor	$394,707,076
Quality Control	$56,930,556
Materials and Equipment	$125,523,948
Engine	$331,039,707
Avionics	$18,562,826
Test Facilities	$0
Subtotal	$3,499,067,132

We can then do the same with the sum of all the production costs to find their subtotal shown in Table 12-13. When dividing this subtotal by the quantity of aircraft being produced, we can get the single unit cost, also shown in Table 12-13.

Table 12-13: Total Production and Unit Cost

Sustaining Engineering	$1,080,190,892
Tooling	$931,914,596
Manufacturing Labor	$5,243,382,776
Quality Control	$699,348,511
Materials and Equipment	$3,110,024,848
Engine	$12,260,729,892
Avionics	$447,958,872
Manufacturing Facilities	$0
Subtotal 500 Aircraft	$23,378,843,316
Single Unit Cost	$46,757,686

If we want to go a step further, we can add both subtotals for DT&E and production to get a final program cost of $26,877,910,448.

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Appendices

Work Breakdown Structure

We submitted the signed formal work breakdown structure in the TM-1 submission. The spreadsheet is far too large to imbed into the document.

The G.1 spreadsheet contains a large amount of the spreadsheets used for the aircraft. Some of these spreadsheets are duplicated below. The different pages are labeld as follows: Wing_Area_Sizing _(Ben), Structal Weight-MatProperties, Mission Weight, WingLayout, Takeoff Calculator, Landing Calculator

G.2 AE460Wing_Layout.xlsx

The Excell sheet G.2 is used to find all of the calculations regarding any wing layout values and dimensions. These were then used to help create the solidworks models of the plane.

G.3 AIAA_Weight_Estimator.xlsx

Sheet G.3 is a calculator for Roskams method of mission weight estimation.

G.4 H_Stab_Layout_Tool.xlsx

G.4 calculates the horizontal stability and graphs the leading edge and trailing edge of the horizontal stabilizers.

G.5 Takeoff_Calculator.xlsx

The G.5 sheet calculates the building velocity, lift, and drag across the runway deck of the carrier.

G.6 V_Stab_Layout_Tool_Fixed.xlsx

G.6 calculates the horizontal stability and graphs the leading edge and trailing edge of the vertical stabilizers.

G.7 Lifecycle Cost Estimation.xlsx

The G.7 sheet is a calculator for all of the values on the DT&E and can give the total program cost and aircraft unit cost.

G.8 totalCMCalcs_460A_11.15.25

Sheet G.8 has multiple different pages and can account for all of the aerodynamic leads calculations.

MATLAB Code

```
%% AE 460A – Section 9.6
% Approach Speed vs Weight (Carrier Landing, WL Configuration)

clc; clear; close all;

%% Landing weights (lb)
W = [22205, 29538, 42567];     % [Min, Design, Max] WL weights

%% Approach speeds (knots) – provided values
Vapp_kt = [124.8788266, 144.0303327, 172.9019703];

%% Plot
figure;
hold on; grid on; box on;

plot(W, Vapp_kt, 'o-', ...
   'LineWidth', 2, ...
   'MarkerSize', 8);

% AIAA approach speed limit
yline(145, '--r', 'LineWidth', 2);

%% Labels and formatting
xlabel('Aircraft Weight, W (lb)', 'FontSize', 12);
ylabel('Approach Speed, V_{app} (kt)', 'FontSize', 12);
title('Carrier Approach Speed vs Weight (WL Configuration)', 'FontSize', 13);

legend('Approach Speed', 'AIAA Limit (145 kt)', ...
   'Location', 'northwest');

%% Annotate points
labels = {'Minimum WL Weight', 'Design Landing Weight', 'Maximum WL Weight'};
for i = 1:length(W)
```

```
    text(W(i), Vapp_kt(i) + 2, labels{i}, ...
        'HorizontalAlignment', 'center', ...
        'FontSize', 10);
end

% --- FLIGHT ENVELOPE GENERATOR
clear; clc; close all;

% 1. AIRCRAFT PARAMETERS
W = 39943;          % Weight (lb)
S = 600;            % Wing Area (ft^2)
CLmax = 0.9;        % Stall CL
CD0_subsonic = 0.04;% Subsonic Parasitic Drag
k = 0.0391;         % Induced Drag Factor
T_SL = 2 * 11000;   % Total Thrust (lbf)

% 2. CONSTANTS & SETUP
g = 32.174; rho0 = 0.002377; R = 1716;
altitudes = linspace(0, 65000, 150);
M_stall_curve = [];
M_thrust_curve = [];
valid_alts = [];

disp('Calculating Flight Envelope...');

% 3. CALCULATION LOOP
for h = altitudes
    % --- Atmosphere Model ---
    if h < 36089
        temp = 518.67 - 0.003566 * h;
        rho = rho0 * (1 - (0.003566 * h) / 518.67) ^ 4.256;
    else
        temp = 389.97;
        rho = 0.000706 * exp(-g * (h - 36089) / (R * temp));
```

```
end
a = sqrt(1.4 * R * temp);

% --- A. STALL LIMIT ---
m_stall = sqrt((2 * W) / (rho * S * CLmax)) / a;

% --- B. THRUST LIMIT (With Wave Drag) ---
sigma = rho / rho0;
T_avail = T_SL * (sigma^0.8);

m_thrust_lim = NaN;

% HIGH RESOLUTION SCAN (Step 0.001)
for m = 0.1:0.001:1.6
    % Wave Drag Correction (Drag rises after M0.8)
    d_cd_wave = 0;
    if m > 0.8
        d_cd_wave = 2.0 * (m - 0.8)^3;
    end

    CD_total = CD0_subsonic + d_cd_wave + k * (W / (0.5 * rho * (m*a)^2 * S))^2;
    Drag = 0.5 * rho * (m*a)^2 * S * CD_total;

    if T_avail >= Drag
        m_thrust_lim = m;
    end
end

% Store valid points
if ~isnan(m_thrust_lim) && m_thrust_lim > m_stall
    M_stall_curve = [M_stall_curve, m_stall];
    M_thrust_curve = [M_thrust_curve, m_thrust_lim];
    valid_alts = [valid_alts, h];
end
```

```
end

% 4. PLOTTING (Refined Legend)
figure('Color','w', 'Position', [100 100 900 600]); hold on; grid on;

% Plot Curves (and save handles 'p' for legend)
p_stall = plot(M_stall_curve, valid_alts, 'r', 'LineWidth', 2);
p_thrust = plot(M_thrust_curve, valid_alts, 'b', 'LineWidth', 2);

% Fill Safe Area
x_fill = [M_stall_curve, fliplr(M_thrust_curve)];
y_fill = [valid_alts, fliplr(valid_alts)];
p_fill = fill(x_fill, y_fill, 'g', 'FaceAlpha', 0.1, 'EdgeColor', 'none');

% Plot Ceiling Dot
[max_h, idx] = max(valid_alts);
p_ceiling = plot(M_stall_curve(idx), max_h, 'ko', 'MarkerFaceColor','k', 'LineWidth', 1.5);

% Text Label on Graph
text(M_stall_curve(idx)+0.02, max_h, ['  Max Alt: ' num2str(round(max_h)) ' ft'],
'FontSize', 10);

% Styling & Corrected Legend
xlabel('Mach Number'); ylabel('Altitude (ft)');
title(['Flight Envelope: ' num2str(W) ' lbs Gross Weight']);
axis([0 1.2 0 65000]);

% THIS FIXES "data1" -> "Absolute Ceiling"
legend([p_stall, p_thrust, p_fill, p_ceiling], ...
    'Stall Limit', ...
    'Thrust Limit (w/ Wave Drag)', ...
    'Operating Envelope', ...
    'Absolute Ceiling', ...
    'Location', 'best');
```